Analytics Report PSR

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Power systems are evolving rapidly, and so are the analytical frameworks required to understand them. This edition of the Analytics Report brings together studies and methodological discussions on some of the questions now shaping the sector: uncertainty representation, storage valuation, transmission flexibility, hydraulic constraints, and the growing role of advanced computational approaches in planning and decision-making. Together, these contributions show how analytical innovation is becoming essential to address increasingly complex energy challenges. We invite you to read on and explore the analytical perspectives presented throughout this edition.

Special Issue

PSR User Meeting 2026

A preview of the next PSR User Meeting and the strategic discussions shaping its agenda, including resilience, flexibility, climate uncertainty, transmission, and advanced analytical workflows.

inBrief

Energy storage applications

Case studies on the role of energy storage in different system contexts, from large interconnected systems to battery sizing and economic assessment in Colombia.

inDepth

Hydraulic constraints and system cost

A technical discussion on how to assess the economic impact of hydraulic operating constraints and support more objective prioritization of flexibility measures

inSight

Planning under uncertainty

Methodological articles on renewable uncertainty, transmission flexibility in New Zealand, and long-term adequacy planning in the Pacific Northwest.

inNovation

PSR in Academic Research

Highlights from PSR's recent academic activity, including conference papers, publications, and research topics at the frontier of energy analytics.

Editorial

Power systems are evolving under a combination of pressures that are reshaping planning practices and the analytical tools required to support them. The expansion of variable renewables, the growing relevance of storage and transmission flexibility, the operational implications of climate variability, and the need to balance reliability, affordability, and decarbonization are all raising the standard for energy analytics.

The ***Analytics Report*** was conceived as a technical publication focused on how these challenges are addressed through robust models, sound methodologies, and practical studies connected to real decisions in planning and operation globally. Its structure reflects that purpose.

A **Special Issue** frames a broader strategic theme. An **inBrief** presents application-oriented studies. An **inSight** examines methodological choices and their implications. An **inDepth** explores one topic in greater technical detail while keeping the discussion connected to its practical consequences. And An **inNovation** section highlights PSR's research activity and ongoing methodological development.

This edition offers a clear illustration of that approach.

The **Special Issue** on the PSR User Meeting 2026 opens the report by highlighting a landscape in which methodology, computation, and strategy are becoming more closely linked. The two **inBrief** articles focus on **energy storage**, showing how its value depends on system context, representation, and the analytical framework used to assess it. The **inSight** section then turns to three issues that are becoming central to modern planning: the importance of explicitly representing **renewable uncertainty in operation planning**, the role of **transmission flexibility** in New Zealand, and the use of uncertainty, dynamic reserves, and tractable workflows in adequacy planning for the Pacific Northwest.

That same perspective is developed further in **inDepth**, which presents a methodology to assess the economic impact of hydraulic operating constraints in Brazil and to support more objective prioritization of flexibility measures. The edition closes with **inNovation**, which highlights PSR's academic and research activity across conferences, publications, and preprints, reinforcing the connection between tool evolution, methodological development, and applied studies.

Taken as a whole, this issue is organized around a common backbone: **uncertainty, flexibility, system representation, and decision support under increasingly complex planning and operating conditions**. Whether the focus is storage, renewable variability, transmission, adequacy, hydraulic constraints, or research activity, the underlying question is the same: how should analytical tools and studies evolve so they remain both rigorous and useful as power systems change?

We hope this issue offers a clear perspective on how PSR approaches that question and invites a closer reading of the articles that follow.

The PSR Analytics Team

Special Issue

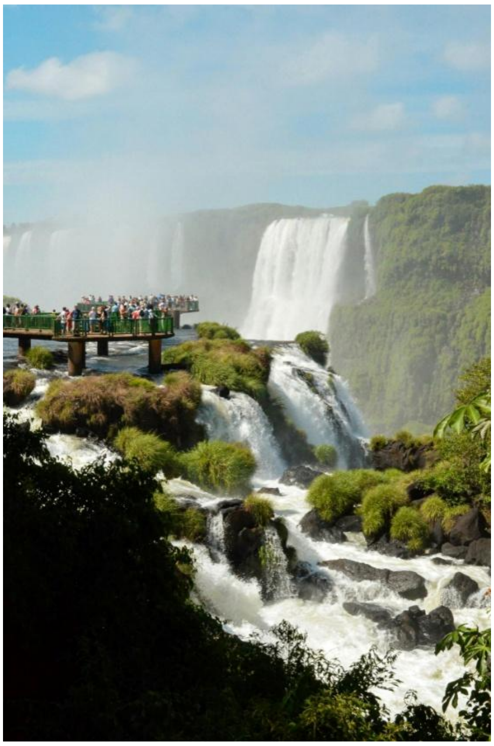
PSR USER MEETING 2026: WHERE THE NEXT QUESTIONS IN ENERGY ANALYTICS TAKE SHAPE

Fernanda Thomé and Lucas Guerreiro

For more than two decades, PSR's User Meeting has served as a forum for technical exchange among professionals engaged in energy planning, system operation, and analytical model development. Over time, however, its role has become broader than that of a conventional user event. As the sector has grown more complex, the meeting has increasingly become a place where methods, tools, and real-world planning challenges are examined together — not as separate conversations, but as different dimensions of the same transformation.

That transformation is now moving quickly. Power systems are being reshaped by the combined effects of decarbonization, electrification, climate uncertainty, network constraints, and rising needs for flexibility. At the same time, the analytical environment itself is changing. Advances in computing power, data pipelines, and artificial intelligence are beginning to influence not only the scale at which problems can be addressed, but also the way studies are formulated, workflows are organized, and decisions are supported. In parallel, the rapid expansion of data centers has become, in its own right, a relevant driver of electricity demand and infrastructure needs, bringing new planning questions to the forefront. In this context, the boundaries between methodology, computation, and sector strategy are becoming increasingly blurred.

The 2025 edition of the PSR User Meeting, held in Búzios, reflected this moment with particular clarity. Bringing together more than 100 participants from 19 countries and 48 companies, along with a large PSR technical delegation, the event offered an international setting for discussions on some of the issues that now define the frontier of energy analytics. Questions related to renewable integration, climate representation, flexibility, transmission expansion, methodological robustness, and practical model application were not treated as isolated topics, but as parts of a broader effort to improve how complex systems are represented and how planning decisions are made under uncertainty. The value of the meeting lay precisely in that combination: methodological depth, practical exchange, and direct interaction between users and developers.



The next edition, to be held in **Foz do Iguaçu, Brazil, from May 25 to 29, 2026**, will continue this trajectory. The setting is particularly appropriate. Beyond its symbolic relevance, Foz do Iguaçu places the discussion close to one of the most emblematic pieces of energy infrastructure in the world: Itaipu. The planned technical visit adds a concrete dimension to the meeting, connecting analytical discussions with the scale, engineering, and operational reality of large power systems.

The 2026 edition is expected to address strategic themes such as resilience and flexibility in power systems, climate change modeling and uncertainty representation, integrated planning of generation, transmission and storage, market design and reliability, advanced optimization methods, and the growing need for scalable analytical workflows supported by high-performance computing, automation, and advanced visualization. Within that broader landscape, it is natural that artificial intelligence will also be part of the conversation, both as a new source of demand through data-center expansion and as an emerging capability that may reshape how studies are prepared, explored, and used by practitioners and decision-makers. Not as a standalone trend, but as one more expression of how quickly the analytical and operational context of the energy sector is evolving.

This is what gives the PSR User Meeting its continued relevance. The event is not important simply because it presents new tools or recent developments, but because it creates a space in which the most pressing technical questions can be discussed before they become settled practice. It is a place to examine what should be represented more carefully, what can now be solved more effectively, what new trade-offs are emerging, and how analytical approaches must adapt to remain useful in a changing sector.

The Special Issue of this new ***Analytics Report*** is therefore a natural place to open the discussion. Many of the topics explored in the pages that follow — from uncertainty representation and storage valuation to transmission flexibility, computational scalability, and research-driven innovation — are closely connected to the same questions that will shape the next PSR User Meeting. We hope to continue that exchange in Foz do Iguaçu, in the company of those working to advance energy analytics in practice.

inBrief

ENERGY STORAGE FOR FLEXIBILITY AND RELIABILITY IN INTERCONNECTED AND ISOLATED SYSTEMS

Jairo Terra, Thais Sobrosa and Fernanda Thompson

The Role of Storage in Energy Transition

The Brazilian power sector is undergoing a profound transformation driven by the rapid expansion of variable renewable energy (VRE) and ambitious decarbonization targets. Although the country's electricity matrix has historically been dominated by hydropower, the accelerated integration of wind and solar has introduced significant operational challenges to the flexibility of the National Interconnected System (SIN). Together, these two sources now account for roughly 40% of total installed capacity, yet the expansion of new hydropower plants, traditionally the system's primary flexibility resource, remains increasingly constrained by social and environmental restrictions.

Notably, approximately 45 GW of VRE originates from distributed generation (DG), which has emerged as one of the most transformative forces in this energy transition. Growing exponentially driven by net-metering incentives, DG has fundamentally reshaped the demand side of the system, deepening the system “flexibility gap”: in 2025, at least 20% of potential clean energy output was curtailed, while the system relied on thermal dispatch to meet peak demand. This paradox underscores the urgent need for flexible resources capable of bridging the gap between a variable supply and a still-inflexible demand.

In contrast to the renewable expansion observed across the SIN, many localities, particularly in the Legal Amazon region and remote rural areas, remain dependent on diesel-based generation. Beyond the environmental toll, these fossil fuel generators impose severe economic burdens on both consumers and public subsidies, with variable generation costs exceeding R$ 1,600/MWh.

In this context, **energy storage emerges as a pivotal technology** to address both dimensions of this dichotomy: on one hand, providing the flexibility needed to accommodate the growing share of variable renewables within the SIN; on the other, enabling reliable and clean energy supply to isolated consumers still burdened by costly and polluting diesel generation.

Energy storage can take many forms, spanning chemical technologies such as batteries, solutions such as pumped hydro, and thermal storage systems. It can equally be deployed across a wide range of configurations, from distributed resources coupled with end consumers to grid-scale assets operating at transmission level, whether integrated with renewable generation in hybrid power plants or operating independently in standalone arrangements.

However, to realize its full potential, this versatile resource requires advancements across all dimensions of the power sector framework, including system planning practices.

Incorporating storage in planning and operation models

Incorporating batteries into energy system models represents a particular challenge, as these assets exhibit a dual operational profile: acting as a load during charging periods and as a generator during discharge. Unlike conventional generation or demand resources, batteries are governed by a specific set of technical and contractual constraints that must be carefully represented to ensure meaningful results. These include maximum and minimum daily cycle limits, maximum dispatch and recharge durations, maximum depth of discharge, and round-trip efficiency losses that affect the net energy balance across each cycle. Capturing these characteristics adequately demands a detailed representation of the system, with high spatial and temporal granularity to reflect the full complexity of battery behavior across different network nodes and operating conditions.

Failing to account for these constraints can lead to suboptimal or even infeasible operational schedules, misrepresenting the true value of storage in the system. In return for this modeling complexity, however, the optimized system gains access to a highly responsive asset capable of reacting to dispatch signals within seconds, offering a degree of operational flexibility that no conventional resource can match.

1. Operational Models

To address this modeling complexity, the SDDP model, a stochastic optimization tool developed by PSR, offers a comprehensive representation of storage assets. The model captures storage capacity, charge and discharge capabilities, efficiencies, and constraints, allowing storage to be co-optimized alongside hydro, thermal, and renewable resources.

Notably, **SDDP's hourly resolution is particularly well suited** to capturing the benefits that batteries bring to the system, reinforcing the need for high temporal granularity discussed. Beyond individual asset representation, SDDP also efficiently represents the operators' ability to re-dispatch batteries along the duration of component failures, such as generating units, transmission lines, and transformers, through its reliability module CORAL. This level of detail makes SDDP a particularly suitable platform for the case studies developed in this work.

2. Planning Models

Beyond its operational value, assessing the role of storage in system expansion planning is equally critical, particularly in understanding how this technology can help preserve Brazil's renewable matrix over the long term. To this end, **storage can be modeled as an expansion candidate in OptGen**, the expansion planning model developed by PSR. Based on Benders decomposition, OptGen identifies the least-cost investment portfolio by balancing capital costs against operational savings. This approach allows storage to compete on equal footing with conventional alternatives, such as thermal power plants, ensuring that the expansion plan reflects the most cost-effective and sustainable pathway for the system while also ensuring revenue adequacy for the selected projects.

The diagram below illustrates this optimization process, in which the decomposition enables iterative communication between the investment module OptGen, responsible for evaluating capital deployment decisions, and the operation module SDDP, which computes the variable operating costs and marginal benefits resulting from each investment configuration proposed by the former. In such settings, despite carrying a still-significant capital cost, storage tends to prevail by virtue of its ability to reduce overall operating costs, enabling more flexible system operation and unlocking the value of otherwise curtailed renewable energy surplus.

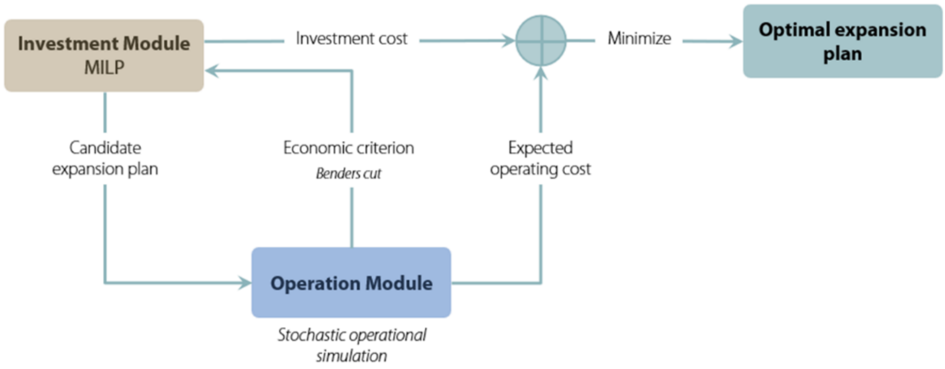


Figure 1 - OptGen optimization scheme

OptGen can also be applied as a hybrid system optimizer for isolated or grid-constrained localities, leveraging its ability to determine the least-cost expansion plan for any system — including small, single-load configurations typical of remote communities. Two modeling features are particularly critical in this context. The first is the representation of demand and generation variability through typical days: by grouping the hours of a given month into representative daily profiles, OptGen substantially reduces the computational burden of the optimization without sacrificing the temporal granularity needed to capture intra-day dynamics such as solar generation ramps and peak demand events. The second is the integration with the Time Series Lab (TSL), PSR's renewable modeling tool, which generates synthetic hourly generation scenarios for wind and solar resources. TSL creates a synthetic hourly generation history by processing data from global reanalysis databases and produces future VRE scenarios that are temporally and spatially correlated with hydrological inflows. To do so, TSL has two main modules: TSL-Data, which builds the synthetic hourly renewable generation record from reanalysis data, and TSL-Scenarios, a statistical model that uses this history to generate future stochastic scenarios while preserving spatial and temporal correlations across all renewable stations. This ensures that the uncertainty in renewable output is properly represented across the many scenarios that OptGen evaluates, enabling robust investment decisions even in small, isolated systems where the consequences of sizing errors are most severe.

2.1 Pumped Storage Candidate Screening with HERA

While models such as SDDP are well suited to represent storage operation and expansion decisions at the system level, the identification of technically and economically viable pumped storage hydro (PSH) sites requires a complementary spatial and engineering tool.

In this context, the HERA software was developed by PSR to evaluate hydropower potential through a structured and computationally efficient process capable of analyzing thousands of alternatives. It supports basin-scale screening and preliminary design by integrating geoprocessing, hydraulic calculations, and cost estimation within a single platform.

For pumped storage applications, the software adopts a bottom-up screening methodology based on successive spatial filters. Starting from a broad area of interest, technical, environmental, regulatory, and topographic constraints are progressively incorporated, reducing the search space and prioritizing the most promising locations.

The methodology relies on high-resolution Digital Elevation Models (DEM) to automatically derive hydraulic head, reservoir geometries, and potential dam alignments. From these inputs, HERA generates schematic engineering layouts, including upper and lower reservoirs, waterways, powerhouse configuration, installed capacity, and preliminary construction cost estimates. The software can assess open and semi-open loop (connected to existing river systems), as well as closed-loop (off-river) configurations. After the filtering phase defines priority areas, an intensive search refines project layouts, optionally optimizing reservoir geometry to minimize barring costs.

By systematically linking territorial screening to preliminary engineering definition, HERA provides a consistent portfolio of technically feasible and economically screened hydropower and pumped storage candidates. These projects can then be incorporated into expansion and operational models, ensuring coherence between spatial feasibility analysis and system-level optimization.

Case Study: Large-Scale Storage Supporting Grid Flexibility

As discussed in the previous section, Brazil's evolving electricity system is increasingly demanding services and capabilities that go beyond what traditional resources can provide. In this study, PSR examines the role of energy storage in delivering flexibility and firm capacity to the National Interconnected System.

The analysis draws on official data from Brazil's Monthly Electric Operation Program (PMO) of February 2025, prepared by the National System Operator (ONS), with a simulation horizon extending through December 2029. All operational simulations were carried out using SDDP, which enabled hourly resolution runs across 400 scenarios — simultaneously capturing the variability of wind, solar, and hydrological inflows throughout the study period.

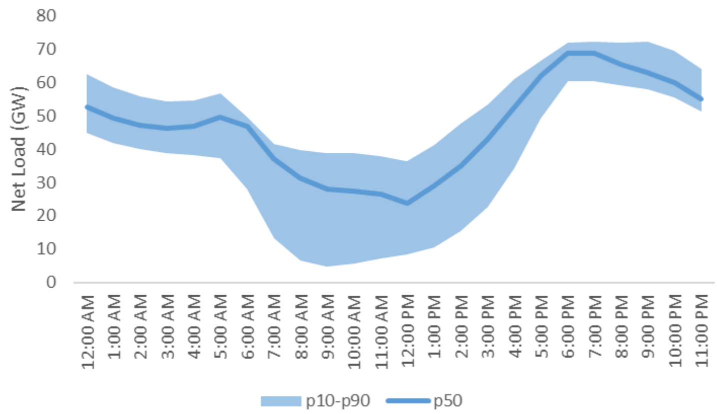


Figure 2 – Average net load in September 2029

To quantify flexibility requirements across different time horizons, PSR assessed the ramp-up needs for 1-hour, 4-hour, and 7-hour windows throughout 2029. Results show that while the annual average 1-hour ramp requirement is around 6 GW, critical scenarios (99th percentile) can push this to 18 GW — nearly triple the average. For 7-hour ramps, critical requirements can exceed 60 GW. These figures can surpass the current combined flexibility capacity of hydropower and thermal plants, which are further constrained by seasonal reservoir levels and operational restrictions.

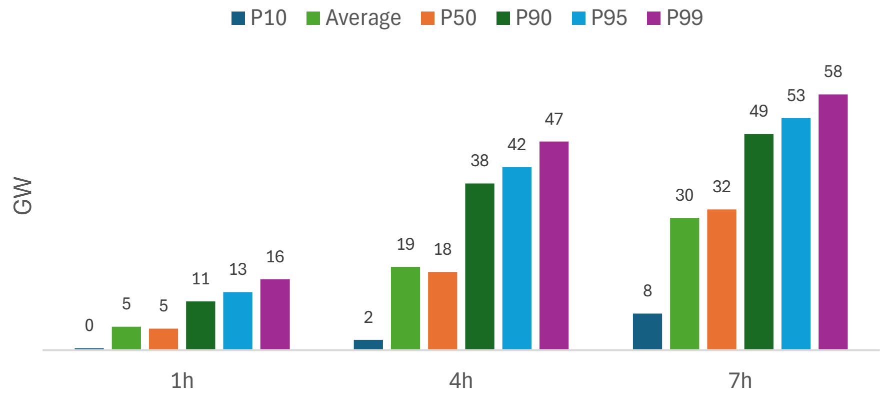


Figure 3 - Required flexibility in the SIN

To address this growing need, the Brazilian government plans to use capacity reserve auctions to contract flexible resources, with gas-fired thermal power plants currently among the main eligible technologies. While a balanced mix of resources is arguably the most robust long-term approach, a direct scenario comparison allows for a clearer evaluation of what each technology contributes — and at what cost — to system flexibility.

For this purpose, PSR assessed four distinct scenarios, with system costs computed by SDDP and comprising operating costs, energy deficit costs[1], and costs associated with violations of operating constraints, primarily water use constraints. Investment costs are not included in this comparison, focusing exclusively on the operational dimension. The scenarios are structured as follows: a Reference Scenario reflecting the current generation mix contracted for 2029, with expired power purchase agreements removed; Scenario B, which adds 32 GW of open-cycle gas-fired thermal plants to the reference configuration; Scenario C, which replaces the thermal addition with an equivalent capacity of short-duration storage in the form of lithium-ion batteries; and Scenario D, which similarly replaces the thermal plants with long-duration pumped-storage hydro. This structure allows storage technologies to be benchmarked directly against the thermal alternative that the government is currently considering for the upcoming auctions.

In all storage cases, the installed ESS capacity was set at 32 GW, distributed across the four submarkets. The costs resulting from all SDDP simulations are summarized in Table 1.

Results show that **both storage technologies outperform the reference** in terms of operating costs: 4-hour batteries reduced average system costs by R$ 1,958 million (13%), while 100-hour PSH delivered savings of R$ 2,298 million (nearly 16%). However, unlike thermal plants, ESS cannot generate energy autonomously when primary resources are scarce — meaning that in energy-deficit scenarios, storage alone is insufficient and must be complemented by firm generation. Taken together, the analyses confirm that **energy storage systems are strong candidates for system expansion**, provided they are combined with dispatchable resources.

[1] As the occurrence of energy deficits is undesirable, a penalty function can be incorporated into the mathematical model of hydrothermal dispatch in the objective function of the problem whenever there is a deficit. With this, the problem of hydrothermal dispatch becomes that of minimizing the operating cost plus the cost of penalizing the energy deficit over the entire planning horizon. Currently, this value is unique, regardless of the depth of the deficit, and is updated annually by Aneel.

Table 1 – Operating costs for expansion scenarios A, B, C and D

Million R$	Total Cost	Operating Cost	Deficit	Others
A – Reference Scenario	14,619	9,233	1,302	4,083
B – Scenario B	14,019	11,405	0	2,613
(B-A)	(600)	2,172	(1,302)	(1,470)
C – Scenario C	12,661	8,845	958	2,858
(C-A)	(1,958)	(388)	(345)	(1,225)
D – Scenario D	12,321	8,764	881	2,676
(D-A)	(2,298)	(469)	(422)	(1,407)

Case Study: Hybrid Systems for Cost-Effective Decarbonization

As industrial and commercial consumers face mounting pressure to reduce their carbon footprint, the question of how to transition away from diesel generation without compromising supply reliability has become increasingly urgent. In this case study, PSR examines the economics and operational performance of hybrid energy systems, combining solar photovoltaic (PV), battery energy storage (BESS), and diesel generation, as a pathway to cost-effective decarbonization for self-producers.

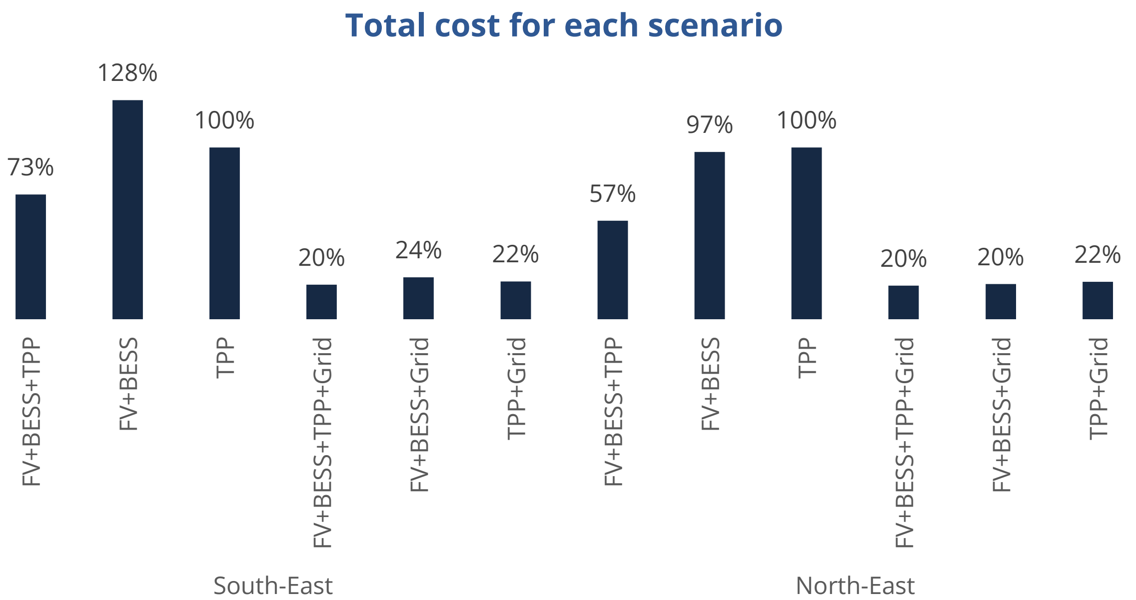
PSR conducted the analysis for three locations in Brazil under two transmission scenarios: with and without grid access. Simulations were performed using PSR's proprietary optimization tools (TSL, SDDP and OptGen) with 7-day typical period representations.

The study evaluates three technology configurations: a conventional diesel-only system (TPP), a fully renewable system combining solar PV and battery storage (PV+BESS), and a hybrid system integrating all three technologies (PV+BESS+TPP). Cost and technical assumptions were defined for each component based on established Brazilian benchmarks. BESS and solar PV were assigned capital and operating costs consistent with current market references, while the existing diesel fleet was treated as a sunk asset, with only variable costs considered. The analysis adopts a 15-year project horizon aligned with the expected BESS lifetime, standard efficiency and performance parameters for batteries and PV modules, and the official 2025 value for the cost of unserved energy.

The main conclusions of the study are summarized as follows:

- Transmission grid access emerges as the single most important determinant of overall system cost and configuration. Grid-connected solutions consistently outperform isolated alternatives across all locations and load levels, highlighting the strong economic value of interconnection.
- Within grid-connected configurations, hybrid systems combining solar PV, BESS, and thermal backup provide the most robust and cost-effective solution, while fully renewable and diesel-plus-grid options also achieve comparable performance when transmission access is available.
- In contrast, isolated systems face a markedly different cost structure. The absence of grid support significantly increases required installed capacity and overall costs, and the economic viability of fully renewable off-grid solutions becomes highly dependent on local solar resource quality and variability.

- BESS plays a pivotal enabling role in hybrid architecture by time-shifting solar generation to evening peak periods, thereby reducing reliance on thermal generation, alleviating grid import constraints during peak hours, and avoiding costly capacity oversizing.
- Geographic differences in solar resource conditions have limited impact in grid-connected scenarios, as the grid effectively smooths variability and equalizes performance across locations. However, in off-grid configurations these differences become decisive, with solar resource availability and consistency emerging as the dominant drivers of techno-economic outcomes.



Conclusions

The analyses presented in this work demonstrate that energy storage is not a single solution but a versatile technology capable of addressing structurally distinct challenges within modern power sectors, from delivering flexibility and reducing curtailment in bulk transmission systems to enabling clean, cost-effective energy supply in isolated communities currently dependent on diesel generation. Across both dimensions, PSR's modeling tools SDDP, OptGen, TSL, and HERA provide the analytical foundation necessary to evaluate, plan, and optimize storage deployment with the rigor that investment and policy decisions demand.

While Brazil offers a particularly compelling case study, the underlying dynamics are far from unique. Whether in emerging economies navigating the tension between rapid VRE expansion and grid adequacy, or in developed nations facing the retirement of dispatchable assets and the decarbonization of remote systems, the core challenge is the same: integrating variable generation while preserving reliability at least cost. The tools and methodologies presented here are designed precisely for this purpose, adaptable to different system configurations, regulatory contexts, and resource endowments, and serve as a replicable framework for any country seeking to maximize the value of storage in its own energy transition.

OPTIMAL SIZING AND ASSESSMENT OF BATTERY ENERGY STORAGE SYSTEMS

Carolina Maldonado and Silvio Binato

The increasing integration of variable renewable energy sources introduces new operational challenges for power systems, particularly in terms of flexibility, reliability, and economic efficiency. In this context, battery energy storage systems (BESS) are becoming a promising technology, as they can contribute to meeting these requirements by providing services such as energy arbitrage, operating reserve, and grid operation support. This article presents a methodology for the optimal sizing and cost-benefit assessment of battery energy storage systems and applies it to a case study of the Colombian power system. The proposed approach is based on the results of stochastic operational simulations, identification of candidate locations, and optimization models to determine the optimal capacity and site of battery storage systems.

Determining the optimal storage capacity

The optimal sizing and siting of battery energy storage systems was structured in **three main stages**:

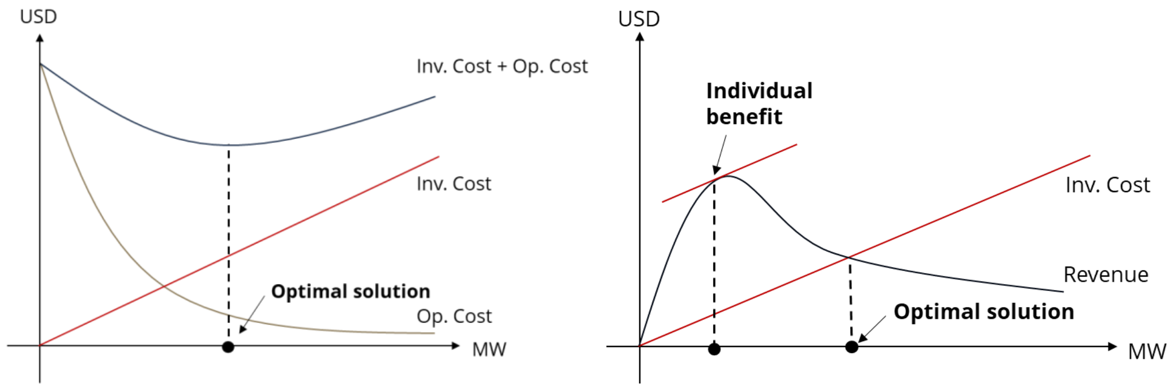
1) **Detailed simulation of the operation** of the power system using the SDDP model to estimate hourly nodal prices and system operating costs, which serve as a reference for the subsequent analyses.

2) **Candidate-node screening**: identification of potential locations for battery installation using a selection methodology based on intraday nodal price differentials. The nodes (transmission system buses represented in detail in the previous simulation) that present the largest intraday price differences correspond to the locations with the highest potential for the installation of energy storage systems.

3) **Capacity and siting optimization**: optimization of storage capacity and selection of the best locations among the candidate nodes identified in the previous step. This optimization is carried out using the OptGen model, which considers candidate battery projects with different storage and power capacities located at the previously identified nodes. The OptGen model evaluates the impact of incorporating these new resources on total system costs by simultaneously considering investment costs and the reduction in operating costs enabled by the integration of storage systems.

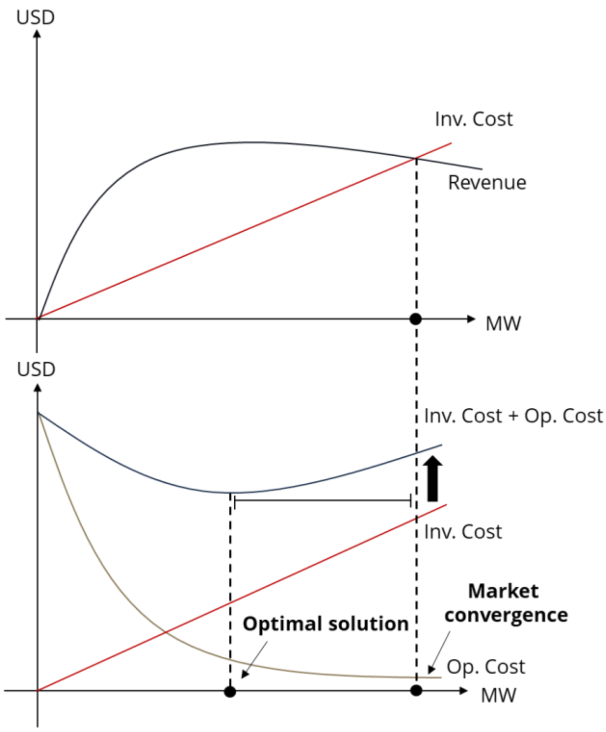
From a centralized planning perspective, the optimal capacity of storage systems is the one that minimizes the total system cost, defined as the sum of investment and operating costs. From an economic perspective, this optimal capacity corresponds to the point at which the derivative of the investment cost for the deployment of storage systems equals the marginal system benefit associated with the reduction of system operating costs, as illustrated in panel (a) of the figure below.

In contrast, in a competitive market context, investment decisions are made by individual agents aiming to maximize their own economic benefits. In this case, the feasibility of the investment depends on the revenues obtained from providing services to the system, such as energy arbitrage and operating reserve provision. Market equilibrium occurs when the expected revenues from storage systems equal the investment costs, as illustrated in panel (b) of the figure below.



a) Centralized Perspective **(b) Market Perspective**

It is important to note that the inclusion of additional remuneration mechanisms, such as fixed capacity or availability payments, may affect market economic signals and lead to investment levels above the system-optimal level. In such cases, the storage capacity resulting from market equilibrium may differ from the level that minimizes the total system cost, as shown in the figure below.



Representation of the distortion in social welfare when fixed availability payments are considered

Can battery energy storage projects be economically viable?

The economic viability of energy storage systems is evaluated through a cost-benefit analysis based on the estimation of revenues obtained from providing services to the system and the investment costs associated with the implementation of batteries.

The main revenue sources considered in the study include: (i) energy arbitrage and (ii) operating reserve provision. Energy arbitrage consists of storing energy during periods of low prices and injecting it into the system when prices are higher, generating revenues from hourly price differentials. Additionally, batteries can participate in the provision of operating reserves, contributing to system reliability and receiving remuneration for their availability and potential activation when required.

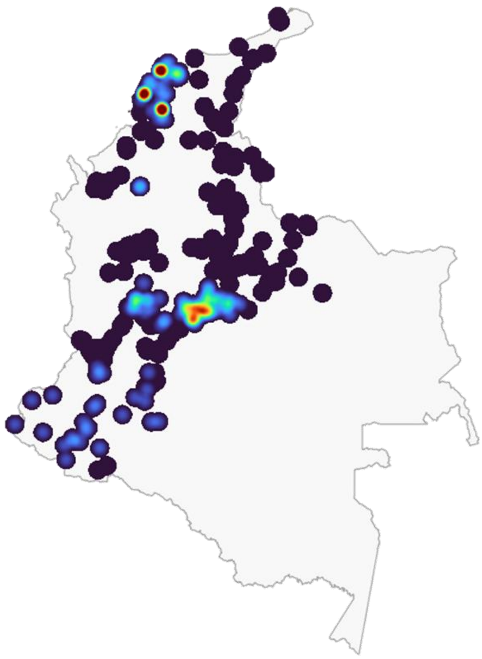
The economic evaluation compares the revenues obtained from these services with the annualized investment costs of the storage systems.

Case Study: Assessing the role of batteries in the Colombian power system

The proposed methodology was applied to the Colombian power system planned for the year 2034. In Step 1, a detailed operational simulation of the system was carried out to estimate marginal operating costs for different nodes of the transmission network.

Initially, several candidate nodes in the transmission network were identified for the installation of battery systems based on the results of intraday price differential analyses of the estimated marginal operating costs at different transmission system buses (as described in Step 2). For each candidate location, different storage configurations were evaluated.

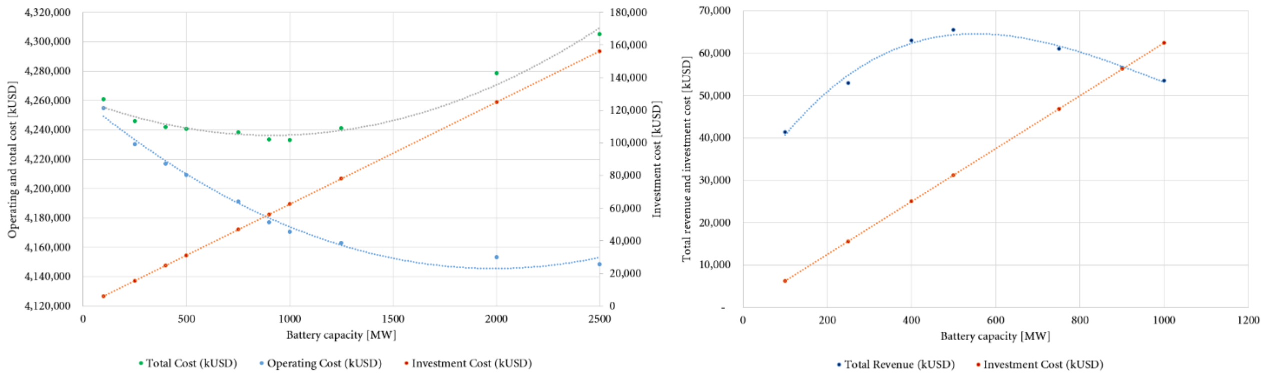
The figure below presents a georeferenced map of the results obtained for the different nodes of the system for a given battery configuration. The results indicate that the most attractive locations are concentrated in the central region and along the northern coast of Colombia. These results highlight the locational value of battery storage systems and how they can provide greater benefits to the system when deployed at strategic locations.



Map of opportunities for batteries

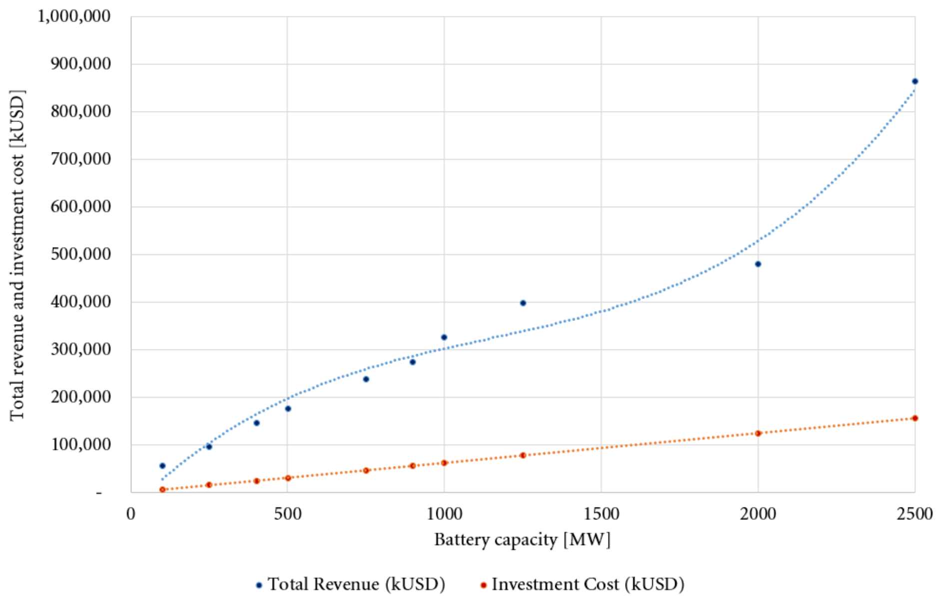
Using the OptGen model, it was possible to determine the optimal level of battery deployment in the system, considering different economic evaluation perspectives. Panel (a) in the figure below presents the system total cost curves as a function of the storage capacity, allowing the identification of the point that minimizes the sum of investment and operating costs. For the conditions analyzed in 2034, the results indicate an **optimal deployment of approximately 900 MW** of storage in the Colombian power system.

Additionally, the results were evaluated from a market perspective, considering the revenues obtained by storage systems through energy arbitrage and operating reserve provision. Panel (b) in the figure below presents the relationship between expected revenues and investment costs for different levels of battery deployment. It can be observed that, under these conditions, the level of storage deployment from the market perspective coincides with that obtained under the centralized planning perspective. That is, the market equilibrium point, where the revenue and investment cost curves intersect, also occurs at approximately 900 MW.



(a) Total costs **b) Total revenues**

In addition, scenarios considering additional remuneration mechanisms were also analyzed. As illustrated in the figure below, these mechanisms may affect market economic signals and lead to investment levels above the system-optimal storage capacity.

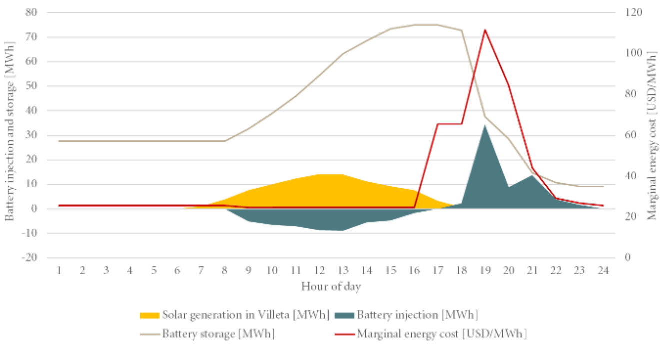


Total revenues for optimal sizing considering fixed availability payments

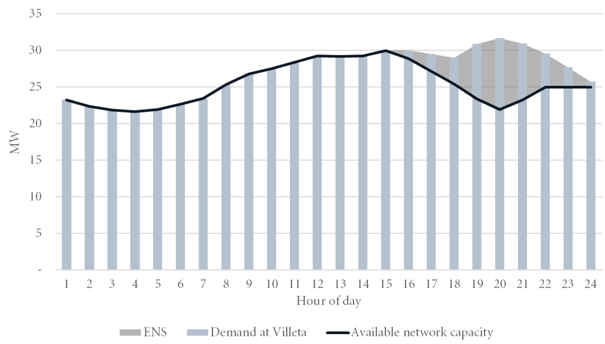
Based on the results of the optimization model, it was also possible to identify the optimal distribution of storage capacity across different points in the grid. The table presents the allocation of batteries at the main substations of the system for the 2034 scenario, highlighting the locations with the highest system benefit for the integration of these resources.

Substation	Voltage (kV)	Battery duration (hours)	Capacity (MW)
Porto Nuevo	110	1	415
Villeta	115	1	75
Cordialidad	110	1	178
San Marcos	110	1	232

Finally, to illustrate the operational impact of batteries on the transmission network, a detailed analysis of the operation of one of the selected locations is presented below. Panel (a) in the figure below presents an example of the hourly operation of the battery installed at the Villeta substation in a day. It can be observed that, during nighttime hours, when there is no solar generation and the available capacity of the transmission lines becomes insufficient to meet the local demand, the battery begins to operate by injecting energy into the system. This discharge helps supply part of the demand during peak-load hours, as shown in Panel (b) in the figure below, reducing the risk of energy not supplied (ENS) and relieving bottlenecks in the transmission network.



(a) Battery operation at Villeta



(b) Representation of ENS at Villeta

From an economic perspective, a cost-benefit analysis was carried out to assess the financial viability of storage projects at the locations selected by the optimal sizing model.

The table below presents the estimated average annual revenue for each storage system considering revenues derived from services provided to the system, such as energy arbitrage and operating reserve provision.

Battery (Location)	Storage Capacity (MWh)	Investment Cost [MUSD/year]	Average Revenue [MUSD/year]
Porto Nuevo	415	25.9	18.2
Villeta	75	4.7	7.0
Cordialidad	178	11.1	16.5
San Marcos	232	14.5	14.9

The results indicate that **storage projects are economically attractive across the analyzed locations**. In addition, the system-wide benefits associated with the aggregated deployment scenario of 900 MW of storage were evaluated, showing a **reduction of approximately 210 million USD per year in system operating costs**, corresponding to a benefit–cost ratio of 3.74.

Conclusions

The methodological framework proposed in this article integrates stochastic simulations of power system operation, candidate location screening methods, and investment optimization models to determine the optimal capacity and location of energy storage resources.

The application of the methodology to the Colombian power system indicated that the integration of batteries can provide significant operational benefits, including reduced operating costs, increased system flexibility, and reduced transmission network constraints. Under the assumptions adopted for 2034, the results indicate an optimal storage deployment of approximately 900 MW, distributed across strategic points in the network.

From an economic perspective, the results show that storage projects may present positive financial indicators at certain points in the network, particularly when different revenue streams associated with services provided to the system are considered. The analysis also shows that additional remuneration mechanisms may influence investment economic signals and modify the economically attractive level of storage deployment.

Overall, the results reinforce the potential role of battery energy storage systems as strategic resources to enhance the flexibility and reliability of the Colombian power system. The proposed methodology can serve as a supporting tool for planning studies and for the design of policies that promote the efficient integration of storage technologies in the power sector.

Further details on the proposed methodology and its application to the Colombian power system are available in the report titled "Energía flexible para el futuro: integración del almacenamiento en el sistema eléctrico colombiano" (Binato et al., 2026)[2].

[2] Binato, S., Bastos, J. P., Maldonado, C., Morais, W., & Xavier, J. (2026). *Energía flexible para el futuro: integración del almacenamiento en el sistema eléctrico colombiano*. Planas, A., Malagón, E., Casas Bautista, C. D., & Cárdenas Valero, J. C. (Eds.). Available at: https://publications.iadb.org/publications/spanish/document/Energia-flexible-para-el-futuro-integracion-del-almacenamiento-en-el-sistema-electrico-colombiano.pdf

inSight

WHY RENEWABLE UNCERTAINTY MUST BE EXPLICITLY REPRESENTED IN OPERATION PLANNING: EVIDENCE FROM BRAZIL

Luiz Carlos da Costa Junior and Hugo Mueller

Introduction

Many power systems around the world are being reshaped by the energy transition. Wind and solar are scaling faster than any other technologies in modern electricity history, changing not only the generation mix but also the nature of system operation. As non-dispatchable resources take a larger share of supply, variability becomes a structural feature of the system rather than an operational exception.

Brazil offers a particularly illustrative example of this shift. For decades, its electricity system was organized around large hydro reservoirs with multi-month regulation capability, complemented by thermal generation. This structure required a planning mindset that was inherently intertemporal: decisions made today about water use have consequences that propagate for months. As a result, Brazilian operation planning has traditionally been built to manage uncertainty primarily through the lens of hydrology — the stochastic behavior of inflows that determines how much water will be available in the future.

That paradigm is now being expanded. Over the last decade, wind and solar generation have grown rapidly and, in several months of the year, already represent a substantial share of supply. In recent years, wind and solar together have accounted for roughly a quarter of Brazil's electricity generation, and official long-term outlooks project this share to keep rising over the next decade. Unlike inflows, whose uncertainty unfolds over seasonal and interannual horizons, renewable uncertainty is characterized by strong short-term volatility and limited controllability. **The implication is not merely "more uncertainty," but a different kind of uncertainty — one that interacts directly with operational flexibility.**

In this evolving context, hydro reservoirs are increasingly valued not only for their energy contribution, but for their ability to balance variability. They become the system's primary flexibility buffer: the resource that compensates for renewable deviations, stabilizes operation in adverse conditions, and reduces reliance on expensive thermal dispatch during stressed periods. This makes the representation of renewable uncertainty in medium- and long-term planning more than a modeling detail. It becomes a determinant of how the system values water, defines dispatch strategies, and projects costs and marginal price signals across time horizons.

This study evaluates, in a controlled and transparent way, how explicitly representing wind and solar uncertainty in the construction of operating policies changes reservoir management, system flexibility, and economic outcomes in a hydro-dominated yet increasingly renewable-intensive power system.

Methodology

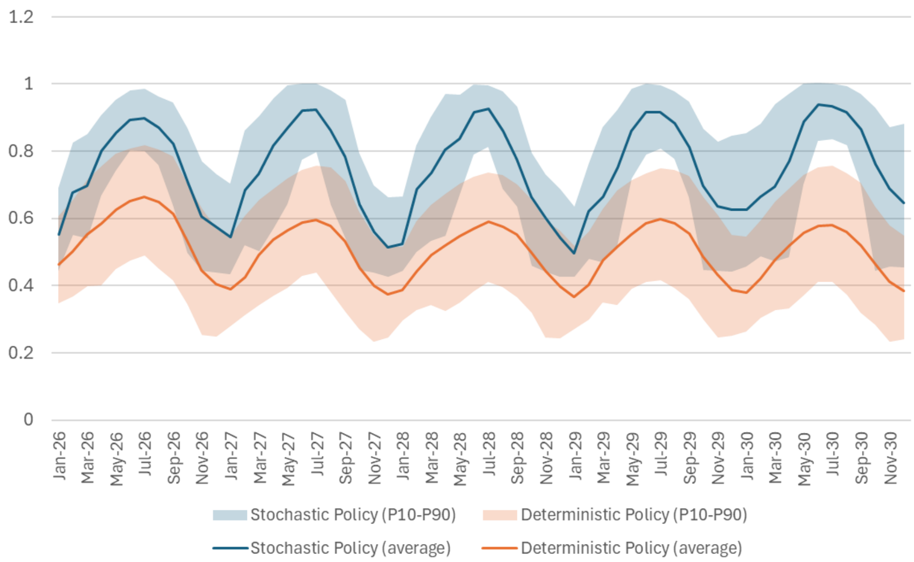
The analysis starts from the official **PMO (Monthly Operational Planning)** dataset for April 2025, published by **Brazil's National System Operator (ONS**). In this framework, wind and solar enter the medium-term planning chain through a deterministic projection, represented as fixed profiles aggregated by subsystem, month, and time block; the same logic applies to other non-dispatchable components built from historical averages. To assess the operational impact of this assumption, we used **PSR's SDDP model** as a single environment to construct operating policies and simulate system operation under uncertainty. Two cases were built from the same PMO reference: (i) a base case with deterministic wind and solar and (ii) a stochastic-renewables case, in which wind and solar variability is represented through synthetic scenarios generated with **PSR's Time Series Lab (TSL)**. TSL reconstructs hourly resource availability for representative wind and solar sites, converts it into multi-year generation scenarios, and aggregates them into the same monthly time-block structure used in the PMO planning chain. The scenarios were bias-corrected so that mean renewable generation matches the official deterministic trajectory, introducing variability around the same expectation. Both cases keep identical assumptions for demand, expansion, operational constraints, and planning structure, isolating the effect of renewable uncertainty representation during policy construction.

In both cases, SDDP first constructs a multi-year operating policy in monthly time blocks (load-duration curve representation) and then evaluates that policy with a higher-resolution stochastic simulation. In this study, the policy was built over 117 monthly stages (04/2025–12/2034) using 1,200 forward sequences and 30 backward openings, and the resulting policies were evaluated through a **five-year hourly simulation** (01/2026–12/2030) with 200 stochastic realizations, **including hydro ramp-up and water travel-time constraints.**

Key results

The comparison isolates a single modeling choice: whether wind and solar variability is internalized during policy construction. The resulting impacts are systematic and coherent across the simulated horizon.

First, **reservoir operation becomes more conservative.** With deterministic renewables in the policy, stored water is implicitly undervalued, leading to more aggressive early releases and lower storage. When renewables are represented stochastically, the policy explicitly anticipates the possibility of weaker renewable output and preserves higher storage levels, increasing flexibility and reducing exposure to adverse realizations.



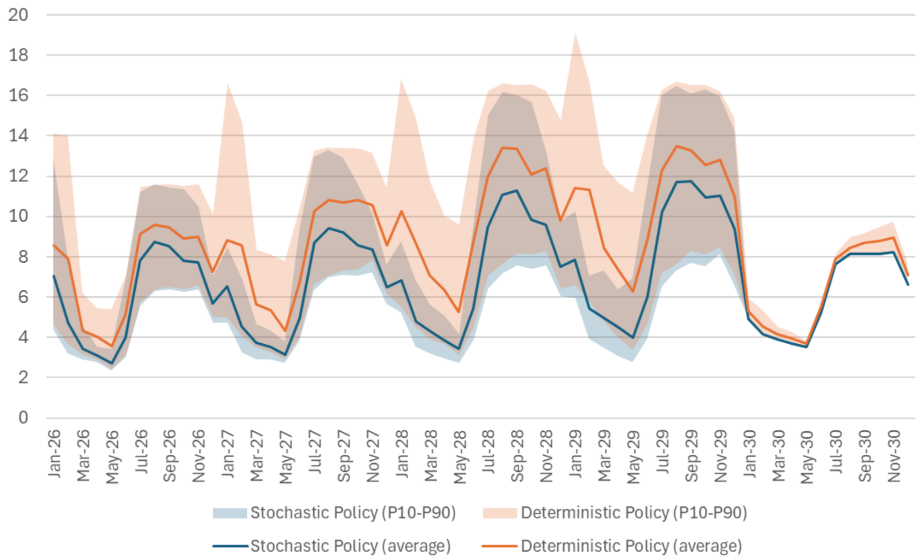
Energy stored in the system (deterministic vs stochastic-renewables policy) – p.u. of maximum storage capacity

Intuitively, when renewables are treated as certain during policy construction, the system behaves as if penalty-related effects will remain close to average levels. **When renewable uncertainty is**

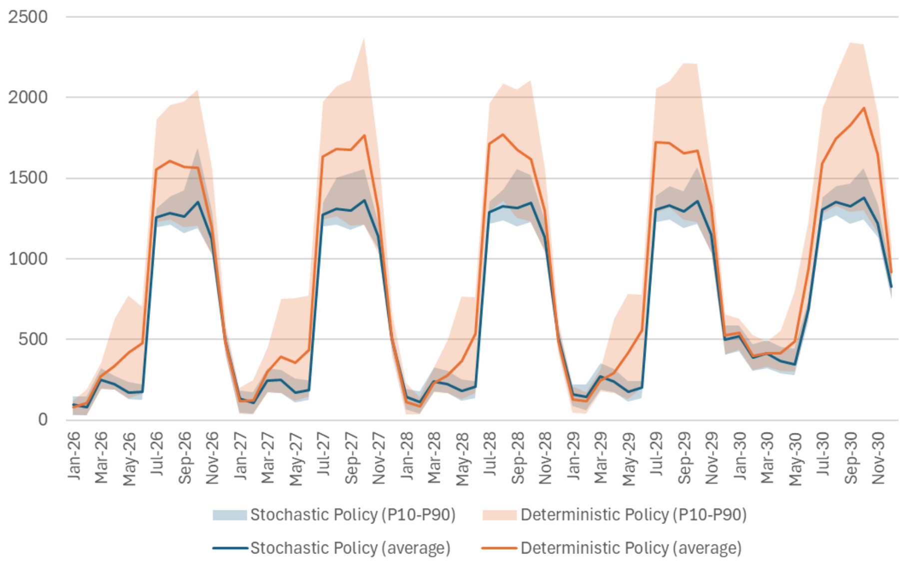
internalized, stored water becomes the natural hedge against low-renewable outcomes, so the policy keeps more optionality in the reservoirs.

This effect is amplified by Brazil's pronounced seasonality. During wet months, higher inflows typically ease dispatch and keep costs lower; during dry months, limited hydro availability increases reliance on thermal generation and raises system costs. A policy that preserves storage therefore creates its greatest value precisely in the dry season, when flexibility is scarce and the thermal stack becomes more binding.

Second, **higher storage translates into lower reliance on thermal generation in stressed periods.** The stochastic-renewables policy reaches dry-season months with more hydro flexibility available, reducing thermal dispatch precisely when thermal costs tend to be highest.

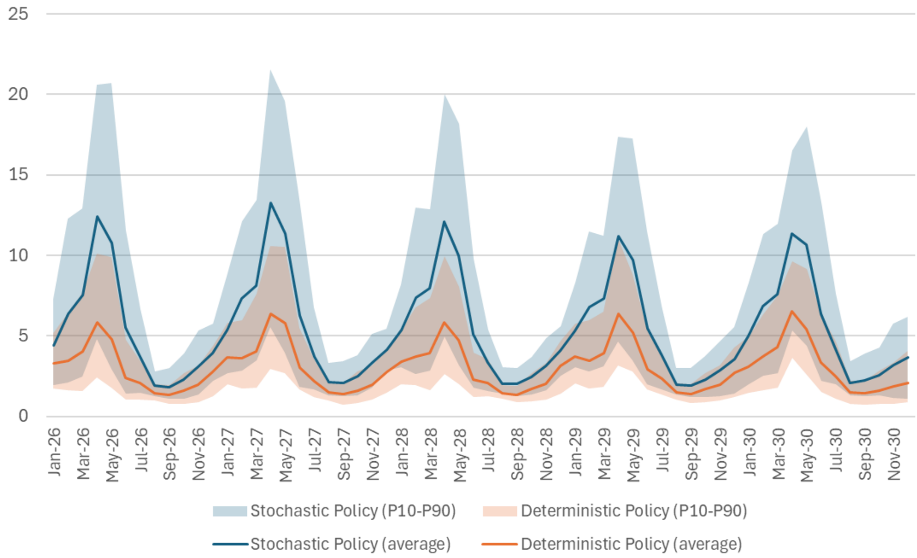


Thermal generation comparison – GW average



Total operating cost over time – M R$

Third, **the trade-off is higher spillage in favorable outcomes**. Operating with fuller reservoirs increases the likelihood of spilling when renewables and inflows realize favorably. In this context, spillage should be interpreted as the operational cost of maintaining optionality: the policy preserves flexibility to protect against adverse outcomes, and in favorable realizations part of that precaution materializes as spill.



Turbinable hydro spillage – GW average

Cost outcomes. Over the simulated horizon, **thermal fuel cost** decreases from R$ 35.066 billion to R$ 33.926 billion, a **3.25% reduction**. The sum of model penalty components decreases from R$ 18.144 billion to R$ 8.030 billion, so **the overall objective (fuel cost plus penalties) declines from R$ 53.209 billion to R$ 41.956 billion (21.15%)**. The penalty changes are directionally consistent with the operational narrative: spillage-related penalties increase, while penalties associated with minimum outflows, irrigation requirements, and minimum operative storage decrease, reflecting the higher-storage strategy and improved flexibility management.

In qualitative terms, the penalty movements follow a clear pattern: spillage-related penalties increase, while penalties linked to minimum total outflows, irrigation requirements, and minimum operative storage decrease; renewable curtailment penalties remain essentially unchanged.

Discussion and conclusions

The behavioral difference between the two policies is straightforward: a deterministic-renewables policy tends to be structurally optimistic, undervaluing stored water and drawing reservoirs down more aggressively, which increases vulnerability to adverse conditions. When renewable uncertainty is explicitly represented during policy construction, the operating logic becomes more cautious, preserves higher storage, and improves the system's ability to absorb unfavorable renewable and inflow realizations using flexible hydro resources.

The study also reinforces a broader planning principle: **risk-aversion mechanisms are most effective when they complement realism, not when they compensate for missing uncertainty.** As planning models incorporate more of the relevant uncertainty structure, prudence tools can focus on truly low-probability, high-impact events instead of correcting for structural simplifications. This distinction matters not only for operational robustness but also for longer-term signals used by market participants and investors. When long-term decisions rely on projections that underrepresent uncertainty, contracting strategies and investment assessments can be anchored on expectations that may later require significant corrections under stressed conditions.

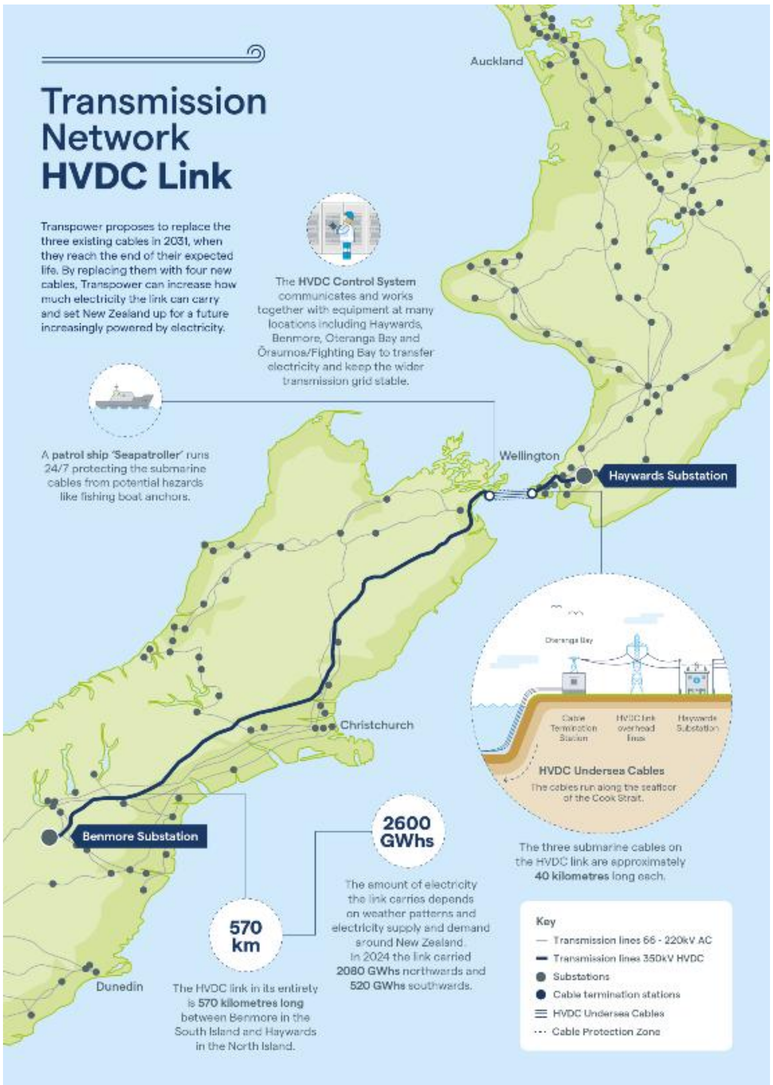
Finally, renewable uncertainty is only one dimension of realism. Additional benefits are expected from improving the realism of the policy construction itself — for example by adopting finer temporal granularity (e.g., hourly), explicitly representing ramping constraints and unit commitment decisions, and modeling transmission with greater detail. These extensions were deliberately left outside the scope of this study: the operating policies were built using the same time-block structure as the PMO reference, while the final evaluation simulation step incorporated only a limited subset of short-term constraints.

QUANTIFYING THE VALUE OF TRANSMISSION FLEXIBILITY: THE HVDC LINK UPGRADE IN NEW ZEALAND

Sean Buchanan and Ricardo Perez

Introduction

Transmission infrastructure plays a central role in enabling modern power systems to operate efficiently under increasing uncertainty. In hydro-dominated systems with geographically concentrated resources, transmission capacity is particularly valuable because it allows energy to move between regions with different supply and demand characteristics.



New Zealand is a clear example of this dynamic. The country's electricity system is divided between the **hydro-rich South Island** and the **demand-dominated North Island**, which relies more heavily on geothermal and thermal generation. The two systems are interconnected by a High Voltage Direct Current (HVDC) link that enables large-scale power transfers between islands.

Transpower, the national transmission system operator, has been evaluating long-term reinforcement options for this interconnection through the **HVDC Link Upgrade Programme**[3]. The analysis presented in "Attachment 6 – Benefits modelling" assesses the economic and operational benefits of upgrading the submarine HVDC cables and associated equipment.

To support this evaluation, the study used a detailed stochastic simulation framework based on **OptGen–SDDP**, enabling the quantification of system benefits across a wide range of future conditions.

This question also reflects ongoing changes in the generation mix. Over the last decade, wind and solar have expanded steadily and, in several months of the year, already represent a non-negligible share of supply. Together, they have recently accounted for about 11% of electricity generation, and official long-

[3] The entire "HVDC Link Upgrade Programme – Major Capex Proposal (Stage 1) – Attachment 6: Benefits modelling" report is available at: https://static.transpower.co.nz/public/uncontrolled_docs/HVDC%20MCP%20-%20Attachment%206%20-%20Benefits%20modelling.pdf

term projections indicate continued growth over the next decade. Unlike hydrological uncertainty, which unfolds over seasonal and interannual horizons, renewable uncertainty is driven by stronger short-term volatility and limited controllability. In this context, hydro reservoirs gain additional importance as sources of operational flexibility, helping absorb renewable variability and limit reliance on costly thermal generation during stressed periods.

The results highlight the critical role of inter-regional transmission capacity in improving system efficiency, supporting renewable integration, and maintaining reliability in hydro-dependent electricity systems.

This study evaluates whether upgrading the HVDC Link would deliver economic and operational benefits to the New Zealand power system from a holistic system perspective.

Modeling Framework: Integrated Planning & Operation Simulation

The benefits assessment combines **generation expansion planning** and **stochastic system operation modeling**.

OptGen is used to determine cost-optimal investment pathways for generation technologies over the long-term planning horizon. This expansion modeling establishes the evolution of the generation mix and ensures that operational simulations are consistent with economically realistic future systems.

SDDP is then used to simulate the operation of the electricity system under uncertainty. The model determines optimal dispatch decisions across thousands of possible future scenarios while explicitly representing hydro reservoir dynamics, renewable variability, and demand uncertainty.

For this study, dispatch simulations were conducted with **hourly temporal resolution over the period 2023–2060**, allowing the model to capture detailed operational patterns and short-term variability in renewable generation and demand.

Hydrological uncertainty plays a particularly important role in the New Zealand system. To represent this variability, SDDP uses **synthetic inflow sequences derived from historical data**, preserving seasonal patterns and correlations between hydro plants.

This stochastic framework enables the evaluation of transmission investments not only under average conditions, but also across a wide range of hydrological outcomes.

Representation of the HVDC Interconnection

The HVDC link between the North and South Islands is explicitly represented in the model through transfer limits, losses, and operational characteristics.

In the current configuration, the interconnection allows transfers of approximately **1,071 MW northward and 762 MW southward**, reflecting operational constraints associated with reactive support equipment at the Haywards converter station.

Following the commissioning of additional reactive support in 2027, the transfer capacity increases to **1,200 MW northward and 950 MW southward**.

The study evaluates three long-term infrastructure strategies:

- **Base Case (Option 1)** – no additional investment; the HVDC eventually loses capacity due to cable failure and is decommissioned in 2038.
- **Option 2** – replacement of the existing submarine cables with identical capacity.

- **Option 3** – upgrade of the interconnection to **1,400 MW northward transfer capability**, enabling larger transfers from the hydro-rich South Island.

These alternatives are evaluated under multiple future scenarios representing different demand, technology, and energy transition pathways.

Modeling Reserves and Reliability

Beyond energy transfers, the HVDC link also plays a critical role in **sharing operating reserves between islands**.

The model explicitly represents reserve requirements associated with the potential loss of large generation units or HVDC poles. In each island, the system must maintain sufficient standby capacity to cover these contingencies using hydro units, thermal plants, batteries, or interruptible load.

When large power transfers occur through the HVDC link, the required reserve margin increases because the potential loss of a pole represents a significant system event. As a result, the modeling framework captures the interaction between transmission capacity, system security requirements, and operational costs.

Key Results

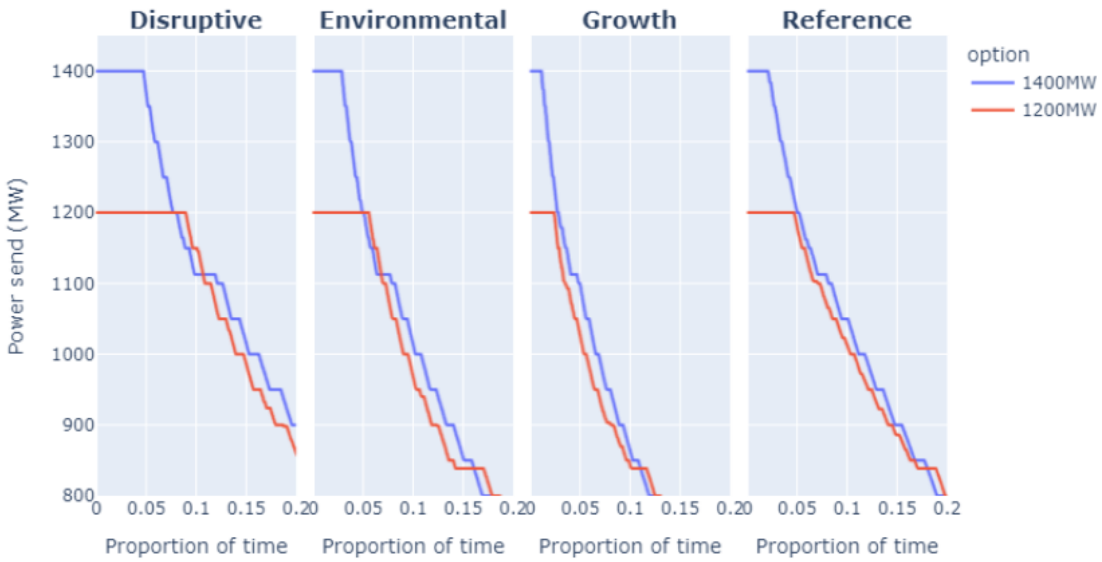
The simulation results demonstrate that upgrading the HVDC link provides substantial economic and operational benefits across a wide range of scenarios[4].

These benefits arise from **three primary mechanisms**: (i) improved utilization of hydro resources; (ii) reduced system operating costs; and (iii) increased system reliability.

Hydropower dominates electricity production in the South Island. However, without sufficient transmission capacity, surplus hydro generation cannot always be transferred to the North Island where demand is concentrated.

The stochastic simulations show that higher transfer capacity leads to more efficient reservoir management and improved inter-island energy balancing, particularly during wet hydrological conditions. The figure presented below illustrates the flow duration curve for HVDC transfers in 2055 comparing Option 2 (1,200 MW) and Option 3 (1,400 MW):

[4] The analysis considers four demand and generation scenarios developed by Transpower based on MBIE's Electricity Demand and Generation Scenarios (EDGS): the Reference scenario reflects a conservative outlook for demand growth; the Growth scenario assumes stronger electrification and higher demand; the Environmental scenario represents a faster transition toward renewable generation and decarbonisation; and the Disruptive scenario explores higher uptake of distributed technologies such as solar, batteries, and electric vehicles.



Flow duration curves for HVDC transfers in 2055 comparing Option 2 (1,200 MW) and Option 3 (1,400 MW)
(Source: [HVDC MCP – Benefits modelling report])

The **gross market benefits** of the HVDC upgrade combine two complementary sources of value: **generation capital benefits** and **operational cost benefits**. Generation capital benefits arise from changes in the optimal generation expansion pathway. With greater transfer capability between the North and South Islands, the system can make better use of existing resources — particularly hydro generation in the South Island — reducing the need for new generation investments elsewhere in the system. In other words, improved transmission capacity can partially substitute for new generation capacity, leading to avoided or deferred capital expenditures.

Operational cost benefits, on the other hand, reflect improvements in system dispatch efficiency. Increased HVDC transfer capability allows the system to utilize lower-cost generation more frequently, reduce reliance on thermal units, and better manage hydro resources under varying hydrological conditions. These improvements translate into reductions in fuel costs, variable operating costs, reserve procurement costs, and the risk of energy deficits. The **table presented below summarizes the total gross market benefits**, which correspond to the combined effect of these investment and operational efficiencies across the evaluated scenarios.

		Disruptive	Environmental	Growth	Reference
Base Case Option 1	Capital cost benefits	-	-	-	-
	Operating costs benefits	-	-	-	-
	Total benefits	-	-	-	-
Option 2	Capital cost benefits	2,269	3,341	2,880	1,973
	Operating costs benefits	2,462	2,467	1,975	1,623
	Total benefits	4,730	5,808	4,854	3,596
Option 3	Capital cost benefits	2,269	3,341	2,880	1,973
	Operating costs benefits	2,668	2,702	2,104	1,765
	Total	4,936	6,043	4,984	3,737

Gross market benefits present value relative to Base Case (option 1) – 2025 $m; 5% discount rate

(Source: HVDC MCP – Benefits modelling report)

Insights for transmission planning

This HVDC case study illustrates how **transmission infrastructure can unlock system-wide value beyond simple congestion relief**.

In hydro-dominated systems such as New Zealand, transmission upgrades enable more efficient utilization of water resources and improve the system's ability to respond to hydrological variability.

The analysis also highlights the importance of considering **operational flexibility and reserve requirements** when evaluating transmission investments. Increased transfer capacity not only affects energy flows but also influences security constraints and reserve procurement needs.

These interactions can only be captured through detailed stochastic modeling frameworks capable of representing both long-term investment decisions and short-term operational dynamics.

PLANNING UNDER UNCERTAINTY: HOW NWPCC IS PREPARING THE PACIFIC NORTHWEST FOR THE ENERGY TRANSITION

Alessandro Soares

The Northwest Power and Conservation Council (NWPCC) is a U.S. interstate agency responsible for developing a regional power plan for the Pacific Northwest. Every few years, the Council produces a comprehensive plan that sets the direction for how the region should meet its future electricity needs over a 20-year horizon, balancing cost, reliability, environmental policy, and risk.

The upcoming Ninth Power Plan represents a major shift in the Council's analytical approach. Recognizing that the power system's growing complexity — driven by increasing variable renewable penetration, evolving clean energy policies, and uncertain demand trajectories — required a more integrated modeling framework, the **NWPCC adopted OptGen and SDDP, developed by PSR, as core tools** for its regional capital expansion analysis.

In a recent System Analysis Advisory Committee (SAAC) meeting held in January 2026, the NWPCC team presented and discussed the full analytical structure being used to develop the plan. This article summarizes how the Council structured its planning problem and the key methodological decisions behind the Ninth Power Plan.

The Northwest planning challenge

The Pacific Northwest power system is dominated by a large hydroelectric fleet, complemented by growing shares of wind and solar generation and a diverse portfolio of thermal, storage, and demand-side resources. Planning this system over a 20-year horizon involves navigating several layers of uncertainty simultaneously: hydrological variability, variable energy resource (VER) availability tied to weather patterns, different demand trajectories reflecting different climate and electrification assumptions, and natural gas price streams.

On top of this, the region faces a rapidly evolving policy landscape. Washington State's Clean Energy Transformation Act (CETA) mandates a transition toward clean electricity, while carbon pricing mechanisms in Washington, California, British Columbia, and Alberta introduce additional cost signals. The Ninth Power Plan must produce a resource strategy that performs well across this wide range of possible futures — not just the most likely one.

The 2021 Power Plan had already identified that the Council's existing models needed to be updated to address these challenges. In particular, the plan called for models capable of dynamically adjusting reserves for different generation mixes and assessing trade-offs between supply-side and demand-side technologies as part of the expansion decision. After evaluating several alternatives, the Council selected OptGen and SDDP as the foundation for its new regional capital expansion framework.

Translating adequacy into expansion signals

Ensuring resource adequacy is central to any power plan. The Ninth Power Plan uses a set of adequacy criteria that goes beyond a simple planning reserve margin. The criteria include frequency-based metrics (loss-of-load expectation values limiting events to 0.1 per season and 0.2 annually), as well as duration, peak, and energy value-at-risk thresholds designed to protect against extreme tail-end events.

To translate these criteria into signals that **OptGen** can use, the Council developed an **Adequacy Reserve Margin (ARM) methodology**. Using GENESYS[5], the Council's adequacy simulation model, the team runs 90 hydrological/load pairings and identifies the capacity needed in each month to satisfy all frequency and severity metrics. These implied capacity needs are then converted into peak and energy reserve margins expressed as percentages above load.

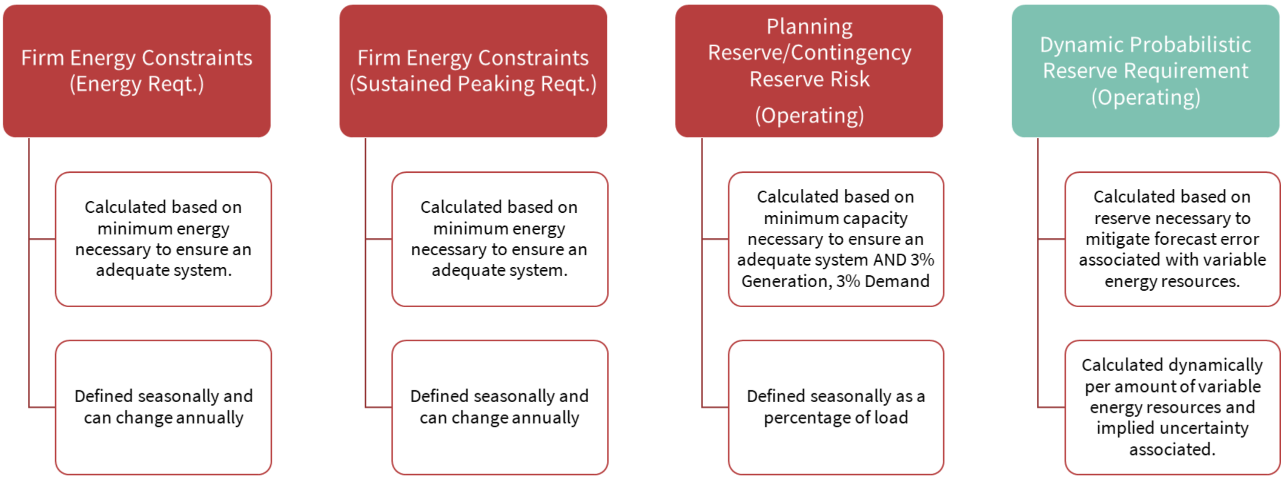


Figure 4 - source: NWPCC SAAC Meeting[6]

The resulting adequacy signals enter OptGen through multiple complementary channels: typical-day load profiles set the hourly demand shape; planning and operating reserves ensure short-term reliability; monthly firm energy constraints signal the need for resources capable of sustained output; and firm peaking requirements capture longer-duration stress conditions. This layered approach ensures that the expansion model receives adequacy information from multiple angles, rather than relying on a single constraint that could be satisfied in an unrealistic way.

As an additional validation step, the Council plans to run OptGen portfolios back through GENESYS to confirm that the resulting resource mixes meet adequacy standards under the full set of simulated conditions.

Dynamic reserves for a changing resource mix

A distinctive aspect of the Council's methodology is the treatment of operating reserves. In a system where the share of variable energy resources is growing, the amount of reserves needed to manage variability and uncertainty is not fixed — it changes with the resource mix itself.

To address this, the NWPCC uses the **Dynamic Probabilistic Reserve (DPR) methodology developed by PSR**, where reserve requirements increase or decrease depending on the forecasted availability of wind and solar generation. Rather than treating reserves as a static exogenous input, the DPR is calculated independently for each reserve planning area (Northwest, California, Mountain West, Southwest, and Canada) based on the variable energy resources within that area. Qualifying reserve resources include existing and new thermal plants, hydro units, and short-, medium-, and long-duration storage.

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[5] Genesys (Generation Evaluation System) is a probabilistic adequacy model originally developed by the NWPCC in 1999 and redeveloped by PSR in 2020, with enhanced capabilities including hourly hydro dispatch, dynamic market pricing, and multi-stage resource commitment

[6] http://nwcouncil.box.com/s/vqpejnpu7tsoqygjloqrdp5e6ytbpcgq

This dynamic representation is particularly important because, as optimization adds more renewables to the system, the reserve requirement adjusts accordingly — creating a feedback loop that more realistically reflects the operational cost of integrating variable generation at scale.

From 13,500 futures to a tractable problem

Perhaps the most consequential methodological challenge in the Ninth Power Plan is managing the huge number of possible futures. Combining all available uncertainty streams — 90 hydro/VER sequences, 15 demand trajectories, 30 gas price paths, and three climate model datasets — **produces over 13,500 unique future combinations**. Running a **stochastic optimization model** across all of them for each year is computationally demanding.

The Council's solution was to select representative futures using a constrained linear programming approach. The team initially explored K-Means clustering but found that random selection from within clusters did not guarantee balanced representation across all uncertainty dimensions — particularly across load pathways and climate model datasets.

The final selection method uses a linear program that applies a set of hard and soft constraints: one future from each 10-percentile bin of net load (annual load minus hydro and VER generation); one from each 10-percentile bin of hydro conditions; a balanced mix of gas prices (four mid-range, three high, three low) with diversified volatility; equal representation of the five load pathways; and a fixed distribution across the three climate scenarios. The objective function minimizes the distance to the population mean within each constraint group.

The Council emphasized that this clustering targets the economic risk space; adequacy signals, which are derived from GENESYS using the full set of 90 hydrological simulations, are not directly affected by the clustering.

Insights for power system planning

The NWPCC's approach to the Ninth Power Plan illustrates how a complex, real-world planning problem can be structured into a tractable analytical process without oversimplifying the system. By combining layered adequacy signals, **dynamic reserves**, and a rigorous uncertainty sampling strategy, the Council has assembled a framework that captures the interactions between investment decisions, operational constraints, and policy requirements.

What stands out in this case is not a single modeling feature, but the **overall design of the analytical workflow** — how each methodological choice was calibrated to address a specific aspect of the planning while keeping the problem computationally manageable. The result is a planning process that can explore a wide range of resource strategies across hundreds of future conditions, providing the Council with the analytical foundation needed to make robust long-term decisions for the Pacific Northwest.

ECONOMIC IMPACT ASSESSMENT OF HYDRAULIC OPERATING CONSTRAINTS IN BRAZILIAN HYDROPOWER PLANTS

Rafael Kelman, Thiago César, Celso Dall'Orto and Pedro Ripper

Context

The Brazilian power matrix is predominantly composed of hydroelectric plants, which are responsible for most of the country's electricity generation. The National Electric System Operator (ONS) coordinates and controls the operation of the National Interconnected System (SIN).

To ensure safe and economic operation, it is essential to consider the operational limitations of these plants, both in real-time operation and in the representation of constraints within electrical-energy models for planning and operation scheduling.

Among the examples of hydroelectric plant variables subject to Hydraulic Operating Constraints (**COPHIs, from the Portuguese acronym**) are water levels (upstream and downstream) and flow rates (outflow, turbined, and spilled), for which minimum and/or maximum value limits may be established, as well as rates of increase or decrease. COPHIs are necessary to ensure:

- Multiple uses of water, compliance with federal and state regulations, and adherence to demands related to socio-environmental issues.
- Adherence to the operating guidelines of each reservoir, which are declared to promote better management of SIN's hydro-energy resources, including definitions from the National Water and Sanitation Agency (ANA) resolutions.
- Safety in the execution of activities and services that require the control of hydraulic variables at the sites.
- The performance of interventions on plant structures (spillways, powerhouses, etc.) that result in some type of restriction on hydraulic variables.

The occurrences of the main upstream and downstream hydraulic operating constraints registered and in effect were observed in the ONS "Hydraulic Restriction Update Request Form" (**FSARH acronym in Portuguese)** extract conducted on November 28, 2024.

	Maximum Water Elevation	**Minimum Water Elevation**	**Water Elevation decrease rate**	**Water Elevation increase rate**
Upstream Constraints	50%	40%	6%	3%

	Max. Water Outflow	Min. Water Outflow	Outflow increase rate	Outflow decrease rate	Ecological minimum Flow	Other
Downstream Constraints	31%	33%	17%	8%	6%	5%

COPHIs are established in Submodule 4.7 of the ONS Grid Procedures ("Inclusion and updating of hydraulic operating constraints for hydroelectric projects"). They are highly relevant to the planning, scheduling, and real-time operation processes, given their implications for hydroelectric plant operation.

Recent years have seen a significant increase in COPHIs registered with the ONS to ensure the multiple uses of water, compliance with federal and state regulations, and adherence to socio-environmental and energy-related demands. In aggregate, these COPHIs reduce the **operational flexibility** of hydroelectric plants, which has economic implications for system operation. For example, it may be necessary to dispatch a gas-fired thermal plant to compensate for short-term variations in renewable sources or to keep it synchronized during periods when hydroelectric plants are constrained.

In 2025, the ONS contracted **PSR**, seeking improvements in the analysis, evaluation, acceptance, and management process of the COPHI constraints that comprise the SIN. One of the activities developed by PSR in this project was a methodology to assess the **economic impact** of each COPHI without judging its necessity. However, by applying the methodology, it is possible to establish a ranking of which COPHIs are the costliest. The expectation is that this exercise will guide more objective discussions on whether it is possible to act to reduce or make these COPHIs more flexible.

A little over a decade ago, during a prolonged drought that particularly affected the São Francisco River, it became clear that it was possible to reduce the downstream flow of some plants, such as the Três Marias HPP. This flow was being used to maintain a specific level required for urban water supply intake. The installation of floating pumps—a simple project with a relatively small budget—allowed for a reduction in these flows, preserving water in the plant's reservoir to prevent it from emptying. This measure provided a benefit far exceeding the cost of the adaptation.

The ranking of the highest-impact COPHIs can help guide where to look first. The Três Marias case illustrates how targeted interventions outside the traditional power-sector scope may yield benefits far greater than their cost, even if there is no guarantee that such success will always be replicated.

This article discusses the project component where PSR develops a methodology for evaluating COPHIs by considering their consequences and repercussions under various operating conditions of the National Interconnected System (SIN). The methodology is grounded in the concept of marginal costs.

Marginal costs

Marginal costs represent the rate of change of the objective function with respect to infinitesimal variations in the independent terms of the constraints (i.e., the Right-Hand Side or RHS of the constraints). In practical terms, they indicate how much the optimal value of the objective function (e.g., total cost or profit) changes when the availability of a resource (such as generation capacity, flow constraints, etc.) is slightly increased or decreased.

In the context of power system optimization problems, marginal costs are widely used to evaluate the impact of constraints. In a hydroelectric plant dispatch problem, for example, the marginal cost of a minimum flow constraint indicates how much the total system cost would increase if the minimum flow were raised. Constraints with high marginal costs are identified as critical to the system and relaxing them can bring significant benefits. Thus, marginal costs are useful for identifying bottlenecks and can guide decisions on where to expand plant capacity or transmission lines.

The marginal cost of the k^{th} COPHI, at a stage t and a hydrological scenario s, expressed by $\pi_{k,t,s}$, measures the variation in the SIN operating cost (in R$) resulting from a change in the COPHI. Some illustrative examples are presented next.

Impact Assessment of Permanent COPHIs

There are currently about 600 active COPHIs, the majority of which are permanent constraints. One possible approach is to evaluate the impact of each COPHI separately by deactivating them one by one. To do this, the difference ΔZ_k is calculated between the operating cost of the SIN simulation considering all active COPHIs (base case) and the cost of the simulation when the k^{th} COPHI is removed.

If the operating cost of the SIN with all COPHIs "deactivated" is Z_0, and if Z is the base case cost includes all COPHIs, then the economic impact of the k^{th} COPHI is calculated as:

$$Impact_k = (Z - Z_0)\frac{\Delta Z_k}{\sum_{j=1}^{K} \Delta Z_j}$$

This cost allocation method, known as "last-addition", or the "first-addition" alternative (which measures the difference between the SIN [National Interconnected System] cost when only the k^{th} COPHI is activated and the cost when no COPHI is active), has the disadvantage of requiring K simulations, which is computationally expensive. Furthermore, it covers only one of the many possible combinations of COPHIs that may be activated or deactivated. Thus, strictly speaking, it would be necessary to evaluate the contribution of each COPHI for all possible combinations of which ones are active.

The Shapley Allocation Method, in turn, is used to distribute costs or benefits among participants fairly, considering each one's contribution to the total result. This method is applied in various fields, such as economics, data science, and resource management. Its basic concepts involve:

- Cooperative Game: when agents cooperate to obtain a collective benefit.
- Characteristic Function (v): represents the value for any subset of players. For a set S of players, $v(S)$ is the value.
- Marginal Contribution: the difference a player i (COPHI) adds when entering a group S, expressed by: $v(S \cup \{i\}) - v(S)$

For a player i, the Shapley value ϕ_i is given by:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)]$$

Where:

- N corresponds to the set of all players.
- S is a subset of participants excluding i.
- $v(S)$ is the value (or cost) of coalition S.
- $v(S \cup \{i\})$ is the value of the coalition formed by adding player i to S.
- $|S|$ corresponds to the size of subset S.

- The fraction $\frac{|S|!(n-|S|-1)!}{n!}$ is the weight given to all formation orders of the groups.

For example, take three power plants (A, B, C) generating energy. The Shapley method calculates how much each plant should be compensated based on its contribution to all possible combinations (A alone, A+B, A+C, A+B+C, etc.).

The practical limitation of applying this method is the number of combinations that would need to be tested for a set of hundreds of COPHIs. To bypass this issue, an extension of this method is suggested: the Aumann-Shapley (A-S) method, usually used in situations where factors are not discrete, but continuous. It is especially useful for cost allocation problems in systems such as power grids or supply chains.

While the Shapley method deals with individual elements, the Aumann-Shapley method deals with continuous fractions of contribution. Instead of allocating costs to individual plants, it allocates costs to shared resources. Its operation is related to:

1. Integration of Marginal Contributions: The method calculates the marginal contribution of each continuous fraction (e.g., each MWh generated) and integrates these contributions along a continuous path.

2. Integration Path: The allocation is made along a path from "no contribution" to "total contribution."

For a continuous resource x, the allocated cost $\phi(x)$ is given by:

$$\phi(x) = \int_0^1 \frac{\partial C(t \cdot x)}{\partial x} dt$$

Where:

- $C(t \cdot x)$ is the total cost when resource x is scaled by a factor t.
- $\frac{\partial C(t \cdot x)}{\partial x}$ is the marginal cost of resource x at point $t \cdot x$.

It is suggested to apply the A-S method to evaluate the impact of COPHIs:

- The system operating cost is the function to be allocated.
- The objective is to quantify the marginal impact of each COPHI on the total operating cost of the SIN, considering their interdependence.

The A-S method allows for fair and efficient allocation. In its discrete fraction version, all COPHIs are jointly scaled from their current value to a minimum value for lower bound constraints ("greater than or equal to") or a maximum value for upper bound constraints ("less than or equal to"). Discrete steps are used to measure the SIN cost increment. The marginal impact of each constraint is obtained via marginal costs. The AS algorithm has four main steps:

Step 1: Define discrete steps

- Choose N (e.g., N =10).
- Define $a_n = \frac{n}{N}$, where $n = 0, \dots, N$ to scale all types of COPHIs, for all plants, stages, and hydrological scenarios. The increment is $\frac{1}{N}$.
    - When $a_n = 1$, the COPHI k receives its original value.

- When $\alpha_n = 0$, the COPHI k is relaxed.

Example of operating flow limits for COPHI *k*:

The premises for the minimum and maximum constraint values must be carefully defined. The relaxed minimum flow could be, for example, the historical minimum daily flow of a plant, while the relaxed maximum flow can be the value used in calculating the flood control volume. This definition is crucial for a more realistic assessment of the economic impact of COPHIs.

Step 2: Simulate the system with tiered constraints

For each step *n* = 0, ..., *N*:

- Scale all COPHIs for all plants by the factor α_n.
- Solve the SIN (National Interconnected System) optimization problem using **SDDP** for medium-term operation planning for these COPHIs that includes different generation technologies (hydropower plants, thermal power, solar power, wind power), the transmission network and storage technologies, such as pumped hydro and battery.
- Calculate the system operating cost z_n and the incremental cost $\Delta z_n = z_n - z_{n-1}$.

Step 3: Calculate the individual contribution of each COPHI k to the cost increment.

Let $\Delta_{k,n} = \sum_{t=1}^{T}\sum_{s=1}^{S}\left(\pi_{k,t,s}^{n} \cdot COPHI_k^n - \pi_{k,t,s}^{n-1} \cdot COPHI_k^{n-1}\right)$ and $\Delta I_{k,n} = \Delta z_n \cdot \frac{\Delta_{k,n}}{\sum_{k=1}^{K}\Delta_{k,n}}$

The final economic impact of each COPHI *k* is $I_k = \sum_{n=0}^{N} \Delta I_{k,n}$

Step 4: Generate the Constraint Ranking

Sort the COPHIs by their impact in absolute values. The ranking should provide a breakdown by COPHI type (e.g., minimum flow, maximum flow, etc.) and by HPP (Hydroelectric Power Plant).

Numerical exercise

The Aumann-Shapley (A-S) procedure for assessing the economic impact of COPHIs was initially tested using SDDP, due to its ease of implementation by PSR and the availability of necessary components (constraint modeling and associated marginal cost preparation). The following assumptions were used:

- May 2025 PMO Deck (Monthly Operation Program), with a 5-year horizon and 200 synthetic inflow scenarios generated by the SDDP's PAR(p) model, without using scenario resampling in the policy.
- A-S was implemented in a Julia code to allocate economic impacts using 10 points (0%, 10%, 20%, ..., 100%), i.e., 11 SDDP runs with policy calculation and simulation for each case using the 200 hydrology scenarios.
- Values for each COPHI were adopted, as shown in Table 1.

Table 1 – COPHI value assumptions for the relaxed case

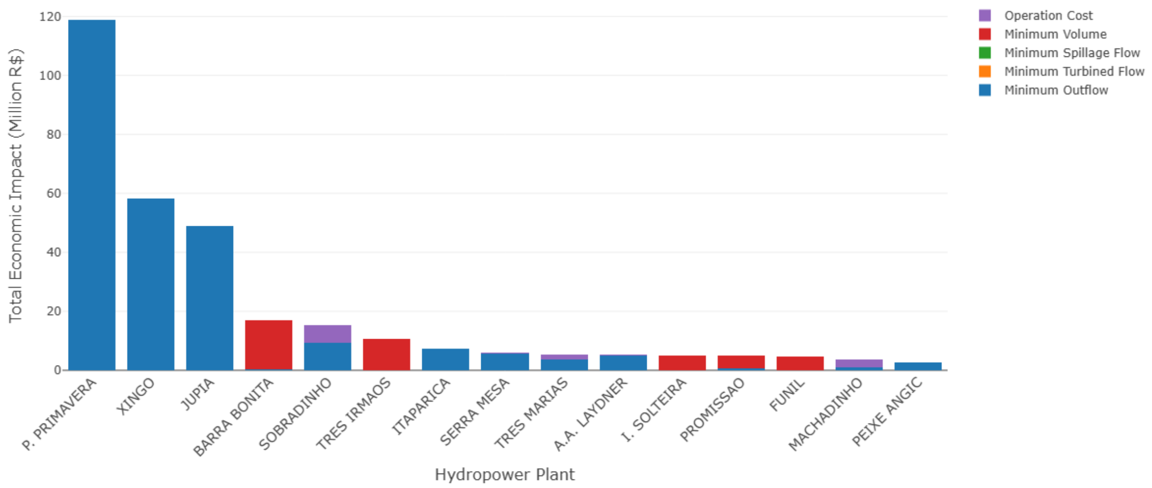
COPHI	Assumption for 100% relaxed case
Minimum outflow	Historical minimum daily total inflow
Maximum outflow	Flow used in flood control volume calculation, if applicable. Otherwise, the minimum between 2x the maximum outflow and the spillway design flow.
Min / max volume	Minimum / maximum operating volume
Min/max water level ramps	Zero
Min/max flow ramps	Zero

Results

The proposed procedure was executed, consisting of 11 SDDP runs (for points ranging from 0% to 100% in 10% steps), the May 2025 PMO deck with a 5-year horizon, 200 hydrological scenarios, and the inclusion (for now) of the previously presented COPHIs:

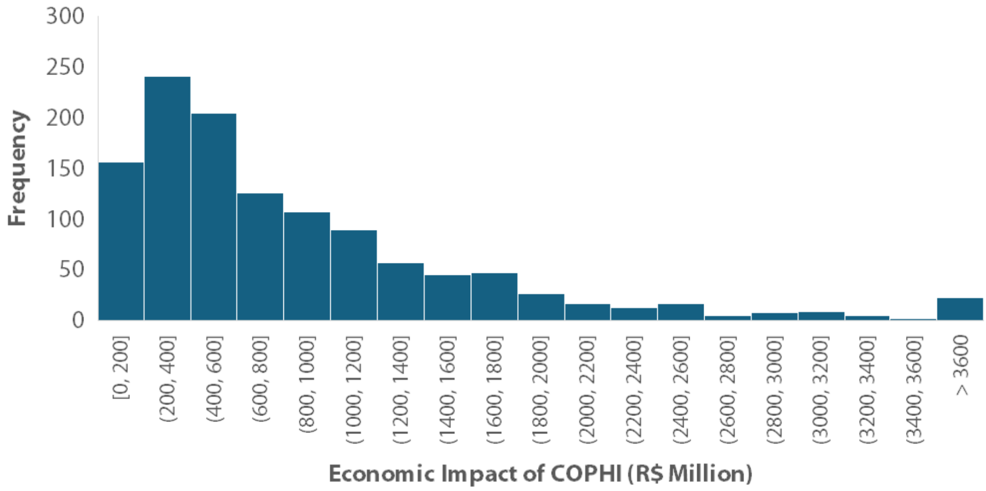
1. Minimum outflows
2. Minimum water level per plant
3. Minimum turbined flow
4. Minimum spilled flow

The following figure presents the allocation of the increase in operating costs in the 5-year horizon in proportion to the product of the marginal cost and the COPHI increment for the 15 plants with the highest values obtained using this methodology.



Among the results presented, it is interesting to highlight the costs of the following plants:

- **Pimental:** presents the highest value (not shown to avoid distorting the others), as it represents the effect of the outflow required to meet the hydrograph; this water is no longer turbined at Belo Monte (a plant with ~90 m of head) and is instead turbined at Pimental (~20 m of head) or spilled. The average impact is R$ 850 million. The following figure shows the economic impact value for each simulated scenario. This helps understand the *dispersion* of the economic impact results. The Pimental site, for instance, has a minimum environmental flow rate (monthly values) that must be kept in the Xingu River, thus does not generate electricity. Depending on the SIN supply condition, this energy loss can have more limited or more severe effects.



- **Porto Primavera and Xingó**: consider their structural minimum outflow values, which are 4,600 m^3/s and 800 m^3/s, respectively.
- **Jupiá**: reflects minimum flow constraints.
- **Barra Bonita, Três Irmãos, and Ilha Solteira**: reflect minimum level constraints for the maintenance of the Tietê-Paraná waterway operations.
- **Sobradinho and Itaparica**: reflect minimum flow constraints and the impact of operating guidelines.
- **Funil (Paraíba do Sul)**: reflects the minimum storage level of 30% defined in Joint Resolution ANA/INEA/DAE/IGAM No. 1382/2015.
- **Machadinho**: reflects the minimum turbined flow conditioned on the reservoir level, as stated in FSAR-H 2858/2022, which defines a minimum flow of 295 m^3/s when the level is above 469.9 m and 250 m^3/s when below that elevation.

Evaluation Criteria and Cost-Benefit Analysis

The ranking of COPHIs by their economic impact, derived from the Aumann-Shapley method, serves as a critical strategic guide. It does more than identify constraints: it helps indicate where the largest potential savings may lie.

While the ranking identifies the costliest constraints, it is important to recognize that "relaxing" a COPHI is not always feasible or appropriate. In practice, the scope for flexibility tends to fall into two categories:

- **Inflexible Constraints**: Some COPHIs are essential for non-negotiable socio-environmental protections. For example, a minimum flow requirement may be vital to ensure the survival of an endangered and endemic species in a specific river reach. In such cases, the "cost" to the ecosystem of relaxing the constraint would be infinite, making flexibility impossible regardless of the economic impact on the power sector.

- **Flexible Constraints via Investment:** Many constraints exist to protect specific human activities that can be adapted with targeted investment. A primary example is the Três Marias HPP, where downstream flow was maintained at a high level solely for urban water supply intake. By investing in floating pumps - a simple, low-budget project - the system allowed for reduced flows and preserved reservoir levels, yielding a benefit that far exceeded the adaptation cost.

To move from a theoretical ranking to practical action, a marginal cost-benefit analysis is required. This involves comparing two distinct curves:

- **The Benefit Curve:** Represents the cumulative reduction in SIN operating costs as a COPHI is incrementally relaxed. This is calculated by simulating the system (e.g., using the SDDP model) under varying constraint limits.
- **The Cost Curve:** Represents the investment required (structural or non-structural) to enable that flexibility.

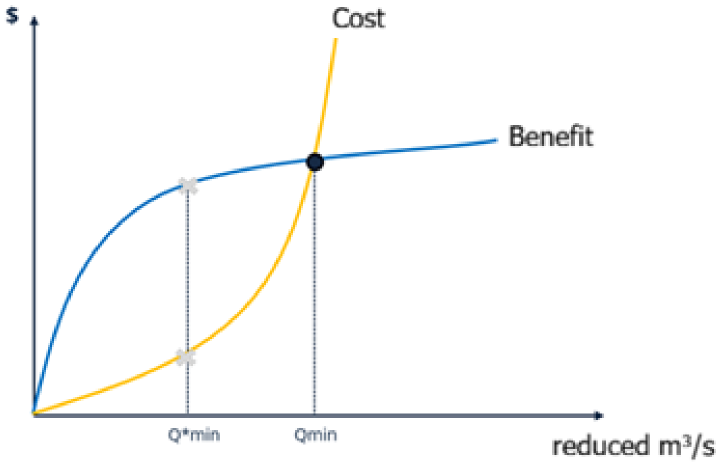


Figure 1 – Cost-benefit curve graph

The optimal degree of flexibility is found at the intersection of these two curves. By applying this logic, COPHIs can be prioritized to guide decision-making considering low-to-moderate cost solutions with rapid or medium-term implementation that provide significant gains. On the other hand, complex structural interventions with high costs and long timeframes may only be justified if the benefit to the SIN is substantial.

This analytical approach is directly relevant to electricity consumers, since any reduction in SIN operating costs ultimately translates into lower tariffs. A significant regulatory hurdle remains, however: the power sector often lacks the legal mandate to fund or execute adaptations outside its immediate domain, even when those interventions would clearly reduce system costs. The Três Marias case illustrates this point well: the installation of floating pumps for urban water intake made it possible to reduce downstream flow requirements, preserve reservoir storage, and generate benefits far greater than the cost of the adaptation. A proactive inter-institutional framework would help enable similar cross-sector investments whenever the benefit to electricity consumers exceeds the cost of the required external intervention.

PSR IN ACADEMIC RESEARCH

Rodrigo Polito and Tiago Andrade

Overview

PSR remains actively engaged in the global research community through publications, conference presentations, and collaborative projects. This article highlights a selection of recent and upcoming academic activities that reflect the company's ongoing work in energy analytics, optimization, and decision-support methodologies.

Last year, **Mario Veiga Pereira, PSR's founder and Chief Innovation Officer**, was included for the fifth consecutive year in the **Stanford-Elsevier Top 2% Scientists ranking**, which identifies researchers with the highest global scientific impact based on publication and citation metrics. More information is available at https://topscinet.com/.

Upcoming Academic Engagements

Upcoming presentations further illustrate the breadth of PSR's current research agenda, spanning uncertainty modeling, decomposition methods, time-series representation, and expansion planning.

XXIV Power Systems Computation Conference (PSCC) 2026

PSR had two papers accepted for PSCC 2026 (June 08-12), in Limassol, Cyprus. The PSR team is set to deliver the following presentations:

- **"Integrated Investment and Operational Planning for Sugarcane-Based Biofuels and Bioelectricity under Market Uncertainty"**, by Carolina Monteiro, Bruno Fanzeres, Rafael Kelman, Raphael Sampaio, Luana Gaspar, Lucas Bacellar, Joaquim Dias Garcia, Bruno Santos.

- **"An Efficient Hybrid Heuristic for the Transmission Expansion Planning under Uncertainties"**, by Joaquim Dias Garcia, Yure Rocha, Teobaldo Bulhões, Anand Subramanian.

2026 SIAM Conference on Optimization (OP26)

PSR had one paper accepted to the 2026 SIAM Conference on Optimization (June 2-5), in Edinburgh, United Kingdom:

- **"General Time Series Approximation in SDDP Using Local Markov Switching Autoregressive Models"**, by Raphael Sampaio, Guilherme Bodin, Joaquim Dias Garcia, Alexandre Street.

Recent Academic Highlights

International Conference on Stochastic Programming (ICSP 2025)

PSR presented five papers at the XVII International Conference on Stochastic Programming (ICSP 2025), held from July 28 to August 1 in Paris, France. Together, these works reflect the diversity of PSR's current research agenda.

1. **"Multicut Benders Decomposition for Generation Expansion with Dynamic Probabilistic Reserves"**, by Tiago Andrade, Alessandro Soares, Fernanda Thomé, Lucas Guerreiro, Luiz Carlos da Costa Junior. This study applies and compares single-cut and multicut Benders decomposition to a generation expansion problem with high renewable penetration and Dynamic Probabilistic Reserve (DPR).

2. **"Incorporation of Risk Metrics Electric System Expansion Planning and Portfolio Efficiency Analysis"**, by Lucas Guerreiro, Alessandro Soares, Tiago Andrade. This study proposes a methodology to incorporate risk measures into energy system expansion planning through a decomposed investment and operation model. By constructing an efficient frontier, it evaluates trade-offs between cost and risk, supporting more robust long-term decision-making.

3. **"ApplicationDrivenLearning.jl: A High-Performance Library for Training Predictive Models in Decision-Making Contexts"**, by Joaquim Dias Garcia, Giovanni Amorim, Alexandre Street. The ApplicationDrivenLearning.jl package in Julia provides a flexible interface and high-performance methods – including exact bilevel formulations and heuristic approaches – to make this technology more accessible.

4. **"Comparison between different parallelization schemes in SDDP policy training",** by Guilherme Bodin, Joaquim Dias Garcia. This study evaluates different parallelization strategies to improve the computational performance of the SDDP algorithm in large-scale hydrothermal dispatch problems. The results indicate that hybrid approaches combining scenario-based parallelization and subproblem-based parallelization provide the most efficient and robust solution for complex energy systems.

5. **"Multicyclic multistage stochastic programming using daisy chains in the electricity market"**, by Gabriel Cunha, Bernardo Costa, Guilherme Bodin, Joaquim Dias Garcia, Gabriel Vidigal, Cleyton Santos. This paper analyzes different time discretization strategies for hydropower scheduling problems modeled with stochastic dynamic programming. Using the open-source IARA tool, it explores how discrete representations approximate the underlying continuous and seasonal dynamics of the system.

Brazilian Operations Research Society Conference (SBPO 2025)

PSR took part in the LVII SBPO (October 5-9), in Gramado, Brazil, with a presentation delivered by its CEO, Luiz Barroso, on *"The Good, the Bad and the Ugly" of Renewable Energy in Brazil and the Challenges They Pose for Operation and Planning*, in addition to three other papers with PSR's participation that were presented at the symposium:

1. **"The IARA Model: A Market Design Simulator for Short-Term Electricity Price Formation"**, by Gabriel Vidigal, Gabriel Cunha, Rafael Benchimol Klausner, Guilherme Bodin, Joaquim Dias Garcia, Carolina Monteiro, Pedro Ripper, Nina Hubner and Bruno Peixoto, showcasing the IARA

software developed through a PSR-led project, coordinated by CCEE and funded by the World Bank. (Further information: https://iara.psr-inc.com/)

2. **"ApplicationDrivenLearning.jl: A High-Performance Julia Package for Training Predictive Models in Decision-Making Contexts"**, by Giovanni Amorim, Alexandre Street and Joaquim Dias Garcia. The ApplicationDrivenLearning.jl package in Julia provides a flexible interface and high-performance methods—including exact bilevel formulations and heuristic approaches—to make this technology more accessible.

3. **"Internalizing the Opportunity Costs of Integrated Operation and the Attributes of Energy Sources in Energy Auctions"**, by Rafael Benchimol Klausner, Alexandre Street, Bernardo Costa, Luiz Barroso and Joaquim Dias Garcia. The Optimized Expansion Auction (LEO) modernizes Brazil's generation planning by replacing traditional auctions with an integrated model that minimizes total costs, ensures reliability, and accounts for interactions among technologies.

Brazilian Seminar on Electrical Energy Generation and Transmission (XXVIII SNPTEE 2025)

PSR was strongly represented at the XXVIII SNPTEE (October 19-22), held in Recife, Brazil, with ten papers presented by team members and participation as co-authors in two additional papers. More information at: https://snptee.com.br/.

1. **"Addressing curtailment: Regulatory Proposals to Improve Efficiency in The Power Sector"**, by Thais Sobrosa, Maria Clara Valente, Maynara Aredes, Jairo Terra.
2. **"Smart Meters in Brazil: Regulatory Approaches and Implementation Strategies"**, by Maria Clara Valente, Thais Sobrosa, Angela Gomes.
3. **"Regulatory and Economic Incentives for Pumped Storage Plants: An International Overview for Application in Brazil"**, by Fernanda Thompson, Thais Sobrosa, Luiz Rodolpho Albuquerque.
4. **"Regulatory Improvements for The Proper Valuation of Hydropower Plant Attributes"**, by Thiago Cesar, Alessandra Daniel, Grazziano Motteran, Ivan Carneiro, Henrique Castro, Celso Dall'Orto, Gisella Siciliano, Fernanda Thompson.
5. **"Optimization of Energy Procurement for Hybrid Wind-Photovoltaic Plant Portfolios in Brazil"**, by Kristian Nolte, Fernanda Thompson, Bruno Fânzeres dos Santos.
6. **"Implementation of a Multiterminal HVDC System in Brazil: Benefits and Results Analysis"**, by Maynara Aredes, André Tiago Queiroz, Rodrigo Martins, Robson Dias.
7. **"A new approach to Physical Guarantee Calculation through an Optimized Energy Auction"**, by Rafael Benchimol Klausner, Alexandre Street, Bernardo Costa, Joaquim Dias Garcia, Luiz Barroso.
8. **"Spatiotemporal Life Cycle Assessment of Hydropower Systems: The Case of Sinop Hydropower Plant"**, by Vanessa Albuquerque, Flora Calili, Luiz Rodolpho Albuquerque, Rafael Kelman.
9. **"The IARA Model: A Mechanism Simulator for Short-Term Electricity Price Formation"**, by Gabriel Cunha, Nina Hubner, Joaquim Dias Garcia, Guilherme Moraes, Gabriel Vidigal, Rafael Benchimol Klausner, Carolina Monteiro, Pedro Ripper.

10. **“Case Study for Methodological Validation of the Valuation of Flexibility, Reliability and Resilience Attributes in a Small-Scale System”**, by Ana Carolina Cunha, Celso Dall’Orto, Renan Pinho.
11. **“Effects of Optimism Bias in Natural Inflow Energy Projections on the Operational Planning of the Brazilian Power System”**, by Arthur Brigatto, Joaquim Dias Garcia.
12. **“ApplicationDrivenLearning.jl: A High-Performance Library for Training Predictive Models in Decision-Making Contexts”**, by Giovanni Amorim, Joaquim Dias Garcia, Alexandre Street.

2025 INFORMS Annual Meeting

PSR presented two papers at the 2025 INFORMS Annual Meeting (October 26-29), in Georgia, United States.

1. **“A Post Processing on Locational Marginal Costs Considering Fixed Losses on DC Optimal Power Flow”**, by Bruno Amaral, Tiago Andrade, Luiz Carlos da Costa Junior, Sérgio Granville.

2. **“Penalty-Free SDDP: Feasibility Cuts for Robust Multi-Stage Stochastic Optimization in Energy Planning”**, by Guilherme Freitas, Luiz Carlos da Costa Junior, Tiago Andrade, Alexandre Street.

Publications and preprints

Beyond conference participation, PSR’s research activity is also reflected in journal publications and preprints addressing topics such as inflow forecasting, life-cycle assessment, expansion planning, and stochastic optimization.

Recent Publications (2025 & 2026)

PSR’s research frequently appears in leading academic journals. A few recent examples include:

- **“Assessing the Optimistic Bias in the Natural Inflow Forecasts: A Call for Model Monitoring in Brazil”.** Arthur Brigatto, Alexandre Street, Cristiano Fernandes, Davi Valladão, Guilherme Bodin, Joaquim Dias Garcia. *IEEE Transactions on Energy Markets, Policy and Regulation*. 2025. This joint work from PSR and PUC-Rio provides empirical evidence of a systematic optimistic bias in Brazil’s official inflow forecasting methodology, based on the PARp-A model, using 14 years of out-of-sample analysis. https://ieeexplore.ieee.org/document/11079762

- **“Integrated Spatiotemporal Life Cycle Assessment Framework for Hydroelectric Power Generation in Brazil”.** Vanessa Cardoso Albuquerque, Rodrigo Flora Calili, Maria Fatima Ludovico de Almeida, Rodolpho Albuquerque, Tarcisio Castro, Rafael Kelman. *Energies*. 2025. This work, part of Vanessa Cardoso Albuquerque’s thesis, validates a spatiotemporal life cycle assessment of the Sinop hydropower plant in Brazil, combining temporal detail and regionalized impacts. https://www.mdpi.com/1996-1073/18/21/5606

- **“Trade-off between computation time and solution quality for integrated generation and transmission expansion planning with N-1 security criterion”**. Lucas Y. Okamura, Carmen L.T. Borges, Gianfranco Chicco. *International Journal of Electrical Power & Energy Systems*. 2025. The paper proposes alternative approaches for integrated generation and transmission planning under the N-1 criterion, balancing solution quality and computational effort in the Chilean system. https://www.sciencedirect.com/science/article/pii/S0142061524006690?via%3Dihub

Preprints

- **"A General and Streamlined Differentiable Optimization Framework".** Andrew. W. Rosemberg, Joaquim Dias Garcia, François Pacaud, Robert B. Parker, Benoît Legat, Kaarthik Sundar, Russell Bent, Pascal Van Hentenryck. https://arxiv.org/abs/2510.25986
- **"An Efficient Hybrid Heuristic for the Transmission Expansion Planning under Uncertainties"**. Yure Rocha, Teobaldo Bulhões, Anand Subramanian, Joaquim Dias Garcia. https://arxiv.org/abs/2602.11490
- **"On the tightness of linear relaxations of alternative mixed integer programming formulations for the generator maintenance scheduling problem"**. Tiago Andrade. https://arxiv.org/abs/2502.08855
- **"Penalty-Free SDDP: Feasibility Cuts for Robust Multi-Stage Stochastic Optimization in Energy Planning"**. Guilherme Freitas, Luiz Carlos da Costa Junior, Tiago Andrade, Alexandre Street. https://arxiv.org/abs/2512.03739